Incremental Land Cover Classification via Soft Label and Subregion Distillation

Bo Ren, *Member, IEEE*, Zhao Wang, Hanyuan Ge, Biao Hou, *Senior Member, IEEE*, Bo Liu, Chen Yang, and Licheng Jiao, *Life Fellow, IEEE*

***Abstract*—With the exponential growth of satellite remote sensing data, land cover classification models must adapt continuously to new classes. However, conventional incremental learning methods face critical challenges: catastrophic forgetting degrades recognition of old classes, and the softmax function further suppresses old-class probabilities due to "class crowding." Existing distillation techniques also struggle to transfer features in irregular geospatial regions. To address these issues, we propose soft labels and subregion distillation (SLSRD). SLSRD mitigates class crowding by employing soft labels instead of hard labels, derived from a hybrid of softmax and sigmoid outputs that preserve richer probabilistic information. Concretely, the soft label is a convex combination of softmax- and sigmoid-based probabilities that preserves interclass relations while relaxing over-confident exclusivity for newly introduced categories, and it supervises all pixels across stages. In parallel, a breadth-first search identifies subregions within each image, which are weighted by probability and size, and similarity between corresponding subregions of the old and new models is maximized. This dual strategy effectively transfers fine-grained knowledge and overcomes the limitations of conventional distillation methods, particularly for large-scale remote sensing imagery. Experiments on three benchmark datasets—Vaihingen, GID, and FBP demonstrate that SLSRD outperforms traditional methods, significantly improving incremental land cover classification.**

***Index Terms*—Incremental learning, land cover classification, soft label, subregion distillation (SRD).**

NOMENCLATURE

Symbol of Stage	
t	Current stage.
$t-1$	Previous stage.
t'	Subsequent stage.
Symbol of Classes	
C^b	Background class.
C^{t-1}	Set of new classes in the $t-1$ training stage.
C^t	Set of new classes in the t training stage.
$C^{t'}$	Set of new classes in the t' training stage.
C	Total set of classes.
c_i	ith class.
Other Main Symbol	
H	Height of the input image.
W	Width of the input image.
N	Total number of pixels of the input image.
$\boldsymbol{X}$	Input image.
$\boldsymbol{Y}$	Input label.
A^{t-1}	Output matrix of the model at the $t-1$ stage.
B^{t-1}	Matrix elements of $\boldsymbol{A}^{t-1}$ are mapped to probability values through the softmax function.
E^{t-1}	Class with the highest probability among the matrix elements representing $\boldsymbol{B}^{t-1}$.
D^{t-1}	Matrix elements of $\boldsymbol{A}^{t-1}$ are mapped to probability values through the sigmoid function.
F^{t-1}	Class with the highest probability among the matrix elements representing $\boldsymbol{D}^{t-1}$.
A^t	Output matrix of the t stage model.
G^{t-1}	Pseudo-label predicted according to the $t-1$ stage model.
L_{c_i}	Probability set of pixel belonging to the c_i in the pseudo-label.
$\tilde{\boldsymbol{L}}_{c_i}$	Decreasing degree of confidence that the pixel belongs to the c_i.
$\boldsymbol{M}$	Pseudo-label generated after the algorithm.

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China, Xidian University, Xi'an 710071, China (e-mail: asdwer2046@126.com; 21171213961@stu.xidian.edu.cn; 24171111327@stu.xidian.edu.cn; avcodec@163.com; 22171214699@stu.xidian.edu.cn; yangchen@xidian.edu.cn; lchjiao@mail.xidian.edu.cn).

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I. INTRODUCTION

IN RECENT years, advancements in satellite sensor technology have facilitated the acquisition of vast volumes of remote sensing data, providing rich spectral information at high spatial resolutions. These high-resolution remote sensing images hold immense application value, as they encompass a broad range, deliver high detail accuracy (Acc), and capture diverse land cover scenes. However, effectively utilizing these resources becomes a crucial challenge. Traditional manual analysis methods are not only time-consuming and labor-intensive but also expensive, with limited Acc guarantees. In contrast, deep learning-based methods for remote sensing image analysis can efficiently leverage these resources and bring significant social value. Therefore, research on deep

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learning-based remote sensing image analysis methods is of great significance.

Semantic segmentation, which assigns a semantic label to each pixel in an image, has become a cornerstone of computer vision. Leveraging fully convolutional networks (FCNs) [1], researchers have extended deep learning from image-level to pixel-level classification, yielding substantial Acc gains. Successive advances-such as multiscale feature representations [2], [3], context modeling [4], [5], and attention mechanisms [6]-have further enhanced segmentation performance. Recently, large-scale vision models have brought transformative capabilities to segmentation. The segment anything model (SAM) [7], for instance, achieves impressive zero-shot performance across diverse domains with minimal prompting. However, despite their generalization ability, these models lack incremental learning mechanisms, making them susceptible to catastrophic forgetting when adapting to new classes or domains. In parallel, integrating large language models (LLMs) into segmentation frameworks-exemplified by LISA [8], LLaVA-Grounding [9], GSVA [10], PixelLM [11], and AnyRef [12]-has improved interpretability and enabled open-vocabulary segmentation through linguistic guidance. Yet most LLM- or VLM-based approaches operate in static or zero-shot modes and have not been adapted to continual learning scenarios. This limitation is especially acute in incremental land cover classification, where the model must retain prior knowledge while accommodating new classes over time.

Beyond these general limitations, semantic segmentation in the remote sensing domain faces unique challenges. Remote sensing imagery often contains massive spatial dimensions, abundant geospatial detail, severe class imbalance, and temporally evolving land cover distributions. These properties exacerbate catastrophic forgetting: without mechanisms to preserve memory for existing classes, model performance on earlier classes deteriorates rapidly after learning new ones [13]. While incremental learning has been widely explored in object recognition [14], [15], [16] and detection [17], [18], [19], research in semantic segmentation-particularly for remote sensing-remains scarce. Furthermore, conventional incremental learning often relies on the softmax function, which can cause "class crowding" when old classes are few in number, leading to further suppression of their predicted probabilities and accelerated forgetting. To address these interconnected challenges, we propose a novel soft label and subregion distillation (SLSRD) strategy. Instead of conventional hard labels, our method employs soft labels for uncertain pixels of old classes, derived from a combination of softmax and sigmoid outputs, which better preserves interclass relationships and mitigates the probability suppression of old classes. In parallel, we introduce a subregion distillation (SRD) mechanism: a breadth-first search identifies all subregions in an image, and their importance is weighted based on predicted probabilities and region size. Similarity between corresponding subregions from the old and new models is then maximized, effectively transferring fine-grained spatial knowledge of old classes to the updated model. This dual strategy enables the network to maintain strong recognition capability for old classes while seamlessly integrating new classes-achieving balanced performance across incremental learning stages in remote sensing semantic segmentation.

During training, SLSRD first applies soft label distillation (SLD) to uncertain pixels belonging to old classes. Each pixel's label is generated by combining softmax and sigmoid outputs, which captures richer interclass relationships than conventional hard labels and better preserves the information of previously learned classes. This design mitigates the probability suppression of old classes, a common cause of catastrophic forgetting in incremental learning. In parallel, SRD is performed to retain spatially localized knowledge. A breadth-first search identifies all subregions in each image, with their importance weighted by predicted probability and region size. For each corresponding subregion pair from the old and new models, similarity is maximized to ensure consistent feature representations. This process transfers fine-grained spatial knowledge from old classes to the updated model, complementing the pixel-level SLD. By jointly applying these two strategies, SLSRD enables the network to integrate new classes while maintaining strong recognition capability for previously learned ones. This leads to balanced performance across incremental learning stages, even in the challenging setting of high-resolution remote sensing imagery with large-scale geospatial data, severe class imbalance, and evolving land cover distributions.

The main contributions of this work are as follows.

1) *Unified Dual-Distillation Framework (SLSRD):* We propose SLSRD, a novel framework for incremental semantic segmentation in remote sensing. By jointly incorporating global soft-label supervision and local subregion alignment, SLSRD effectively mitigates catastrophic forgetting while adapting to newly emerging land-cover categories.
2) *SLD:* We design an SLD mechanism that constructs hybrid soft labels from softmax and sigmoid outputs. This preserves interclass relationships and alleviates the "class crowding" effect, enabling more balanced knowledge transfer across old and new classes.
3) *Subregion Distillation (SRD):* We introduce SRD, a breadth first search (BFS)-based region-aware distillation strategy that decomposes feature maps into probabilistically weighted subregions. By enforcing similarity between corresponding regions of the old and new models, SRD transfers fine-grained spatial knowledge that conventional global distillation fails to capture.
4) *Comprehensive Experimental Validation:* Extensive experiments on three large-scale benchmarks (Vaihingen, GID, and FBP) demonstrate that SLSRD achieves state-of-the-art performance. Compared with existing baselines, it better preserves Acc on old classes, effectively learns new ones, and maintains robustness across different incremental scenarios.

II. Related Work

A. Semantic Segmentation

Semantic segmentation techniques can be divided into conventional methods and those based on deep learning. Con-

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ventional methods rely on feature extraction algorithms and classifier design to differentiate various land covers. However, remote sensing images typically feature a variety of object types, and similar objects may exhibit significant differences in appearance. These challenges make it difficult to obtain satisfactory outcomes with traditional approaches.

Since the development of deep learning, semantic segmentation has made significant progress. In 2014, Long et al. [1] proposed the concept of an FCN, which improved the structure of the original convolutional neural network (CNN). By using a deconvolution layer, it is possible to upsample the feature map generated by the last convolution layer from any size input image. During upsampling, the feature map is restored to the same size as the input image, allowing a prediction to be generated for each pixel while preserving the spatial information. The upsampled feature map is then classified pixel-by-pixel. The latter improvements focus primarily on upsampling and skipping layers of FCN, such as SegNet, DeconvNet, and DeepLab (DL) [20], [21], [22]. U-net [23] is a symmetric semantic segmentation model. Following this, a number of multiscale methods were developed, including DeepLabv3 [24] and PSPNet [25]. Due to different representations and sensor-induced scale transformations, these methods effectively address the issue of different scales of similar objects.

The use of these networks and their variants is widespread in the field of remote sensing. For example, [26] embedded an adversarial complementary learning strategy into a CNN, which is able to extract complementary information from multisource data. In order to extract meaningful multiscale information and fuse features from multisource data, [27] developed an interactive multiscale information extraction block and a global dependence fusion module. In [28], a graph feature extraction module is designed. It aims to extract topological structure information from remote sensing images. In addition, a novel graph fusion strategy called graph dependence fusion is developed. This strategy combines the extracted topological information with rich spectral–spatial information. It also enhances the association and interaction between different graph features. The methods mentioned above represent various algorithms. These algorithms are used for analyzing remote sensing images. They contribute significantly to land cover classification tasks.

B. Incremental Learning

Incremental learning aims to address the problem of catastrophic forgetting, which has been thoroughly researched, particularly in image classification tasks. Previous studies can be divided into three main categories: replay-based, parameter-isolation-based, and regularization-based methods. Each of these approaches offers a distinct strategy for mitigating catastrophic forgetting.

1) Replay-Based: Replay-based methods involve storing some samples of previous training data or generating synthetic samples from previous training data, so that the model can access the old data while processing new data. There are a number of methods that are representative of this concept, including dynamic generative memory (DGM) [29], FearNet [30], ILCAN [31], GAN-Memory [32], semantic segmentation with unknown label (SSUL) with memory [33], adaptive logit regularizer and feature replay (ALIFE) [34], and automatic memory sample selection (AMSS) [35].

DGM employs binary masks to allocate task-specific parameters to overcome interference between tasks and adopts a scalable network to ensure scalability. If pretraining is available, it can provide a relatively stable and strong reference model for continual learning. FearNet and ILCAN also retain statistical information of old features obtained from a pretrained feature extractor, while GAN-Memory continuously adjusts a pretrained generative model with task-specific parameters. For the SSUL algorithm, it stores a portion of data from old classes and defines unknown classes to mitigate background shift issues. The ALIFE algorithm first conducts a thorough analysis of calibration techniques and introduces an adaptive logit regularizer, which allows the model to better learn new classes while retaining knowledge of previous classes. Since the aforementioned methods either randomly select memory samples or use manually designed strategies based on single-factor driving, they cannot guarantee optimality. Therefore, AMSS achieves effective replay by considering comprehensive factors such as sample diversity and class performance.

2) Parameter-Isolation-Based: In general, a method based on parameter isolation extends the network. Its representative methods are PackNet [36], Piggyback [37], and progressive neural networks (PNNs) [38].

By performing iterative pruning and network retraining, PackNet sequentially incorporates multiple tasks into a single network while minimizing performance degradation and storage overhead. Piggyback is a novel method of obtaining good performance on new tasks by taking advantage of the fixed weights of the network. Through the use of a new sensitivity metric, PNNs can leverage prior knowledge by connecting laterally to previously learned features.

3) Regularization-Based: Prior-centric representative methods in the regularization-based approach include path integral (PI) [39], elastic weight consolidation (EWC) [40], and Riemannian walks (RWs) [41]. Knowledge is defined as parameter values that limit the learning of new tasks by penalizing important parameter changes from the old tasks.

The PI introduces intelligent synapses that accumulate task-relevant information over a period of time. In order to memorize learned tasks, the EWC slows down the learning of weights that are critical to those tasks. RW presents RWalk, a generalization of EWC and PI with a theoretically grounded Kullback–Leibler-divergence-based perspective.

Representative data-centric methods are learning without forgetting (LWF) [14], LwF multiclass (LWF-MC) [15], incremental learning techniques (ILTs) [42], modeling the background (MiB) [43], Pseudo-labeling and local pod (PLOP) [44], and repulsion-attraction of sparse and disentangled latent representations (SDRs) [45]. To prevent catastrophic forgetting, these methods aim to retain knowledge from old networks. They utilize knowledge distillation (KD) as a key technique. In addition, they incorporate the distance between activations of the old and new networks. This distance serves

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as a regularization term in the learning process. The regularization helps maintain stability in the model's performance on previously learned tasks.

The LWF can be viewed as a hybrid of KD and fine-tuning (FT). Based on the training data, the learned parameters are discriminative for the new task while maintaining the output of the original task. LWF-MC learns strong classifiers and a data representation simultaneously. ILT retains the knowledge of previously learned classes while updating the current model to learn new ones based on the distillation of the knowledge of the previous model. MiB introduces a new distillation-based framework and introduces a new method for initializing classifier parameters. PLOP proposes a multiscale pooling distillation scheme that preserves long-range and short-range spatial relationships at the feature level. In addition, a pseudo-label based on entropy is designed. SDR employs three methods, including prototype matching, sparse representation, and contrastive learning, to accomplish incremental learning tasks. Prototype matching constructs latent representations to facilitate the model's memory of old classes, sparse representation creates more space for new classes, and contrastive learning enhances the model's discriminative power between old and new classes.

C. Incremental Learning in Remotely Sensed Feature Categorization

Incremental learning aims to address the problem of catastrophic forgetting, which has been thoroughly researched, particularly in image classification tasks. Previous studies can be divided into three main categories: replay-based, parameter-isolation-based, and regularization-based methods. Each of these approaches offers a distinct strategy for mitigating catastrophic forgetting. To address this limitation, incremental learning has been introduced. It enables models to adapt to evolving data distributions. At the same time, it helps retain knowledge of previously learned data. Shan et al. [46] proposed an effective class-incremental segmentation method that mitigates catastrophic forgetting without the need to store old data. Their approach leverages deep feature distillation and label mixing modules, resulting in significant performance improvements on aerial image datasets. Huang et al. [47] tackled the challenge of capturing contextual information by constructing a graph space reasoning module and a dual-channel cross-attention upsampling module, within a new domain-incremental learning framework. This framework allows for better integration of high-level abstract features with low-level spatial features. In the realm of few-shot class-incremental learning (FSCIL), Tai et al. [48] introduced the concept of complete FSCIL (C-FSCIL), where both base and novel class samples are scarce. To address the challenges of distance measurement and catastrophic forgetting, they proposed the mine-distill-prototypes (MDPs) algorithm, which includes a prototypes-distillation network, a prototypes-weight network, and a prototypes-selection training strategy. This method demonstrated superior performance across three different datasets. To alleviate the computational burden on models, Lu et al. [49] introduced a lightweight incremental learning (LIL) approach for remote sensing image scene classification. Their approach makes two key improvements. First, it replaces the traditional feature extractor with a task-common feature extractor (TCE). Second, it introduces a light feature transfer module (FTM). This module helps align data distributions across different tasks. In addition, they introduced a dual-constraint loss function that combines KD and adversarial learning, promoting alignment at both the feature and semantic levels. This approach achieved state-of-the-art performance while minimizing the increase in model parameters.

In conclusion, incremental learning for classifying remote sensing images has progressed significantly in various important aspects. These aspects encompass class-incremental learning, domain-incremental learning, FSCIL, and LIL. Each method tackles particular issues within the field. The research not only confronts challenges like managing both new and existing categories, adapting across different domains, and dealing with limited samples, but also offers solutions that have demonstrated encouraging outcomes and potential for practical use.

D. Incremental Learning for Land Cover Classification

In recent years, deep learning techniques for classification have significantly enhanced both the effectiveness of classification tasks and their applicability. However, these models are prone to catastrophic forgetting when faced with new data and classes that are constantly emerging from around the world. Remote sensing images, in particular, are larger than typical images and contain more information and classes. A single remote sensing image can encompass various classes. Overall, class imbalances are a more critical issue in remote sensing images compared to regular images. In addition, the background classes tend to occupy larger areas in remote sensing images, making the background offset problem more difficult to address.

There has also been research on incremental learning in remote sensing. Bruzzone and Prieto [50] proposed a radial basis function (RBF). This method allows new information to be acquired periodically. It does so as new training sets become available. At the same time, it retains the knowledge gained from previous training sets. During each retraining stage, the network architecture is automatically updated in order to accommodate new classes. A frozen copy of a previously trained network is maintained by Tasar et al. [51] in order to update the network in the absence of preclass annotations. An updated network balances the difference between the outputs from memory and previous classes. Rong et al. [52] propose a feature global awareness module and a label reconstruction module. The former enables the current model to pay more attention to regions related to old classes identified by historical information when learning new classes. During this time, the latter retrieves pixels belonging to the learned class from the background to address the background shift problem and maintain the performance of the old classes.

It is evident that incremental learning in land cover classification is still relatively rare. Our research expands the research content of incremental learning and studies the problems existing in the process of incremental learning of remote

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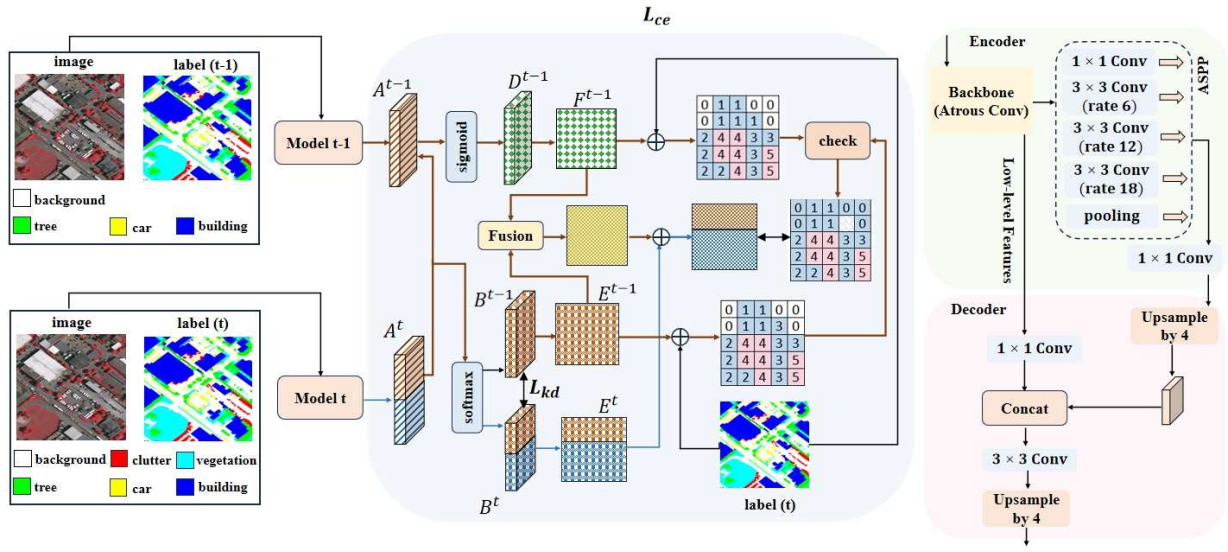


Fig. 1. SLSRD algorithm structure illustration. The left section illustrates the training process, where the model learns a certain number of new classes (green, yellow, and blue) at each incremental stage, while merging the previously learned classes (red and cyan) into the background class. During testing, the model t is used to test all previously seen classes. The middle section shows the prediction inconsistency between the sigmoid and softmax functions. For pixels where the model is not confident (i.e., the predictions differ significantly), these pixels are set to 255, indicating that they are ignored and will not participate in the computation. The right section displays the DeepLabV3 architecture used in this work.

sensing images. Detailed information about the method is provided in Section III.

III. Method

A. Problem Definition and Notation

For the convenience of readers, we have added a symbol table that contains the main symbols that appear in this article. Nomenclature presents the main symbols.

In the context of incremental learning, the current training phase is denoted as t, the previous phase as $t-1$, and subsequent phases as t'. The relationship between the number of classes at different stages is as follows:

$$C = C^b \cup C^{t-1} \cup C^t \cup C^{t'} \tag{1}$$

in which C represents the total set of classes, and C^b stands for background class. C^{t-1}, C^t, and $C^{t'}$ are the set of new classes in the $t-1$, t, and t' training stage, respectively.

The overall frame diagram is shown in Fig. 1. GM stands for the generate merge module. This module generates a trainable pseudo-label map from the merged label, which contains the old classes and the real label of the current stage. This algorithm filters the labels that contain old classes based on the predictions of the previous stage. The cross-entropy (CE) module uses the generated merged label ($\boldsymbol{M}$) and the output matrix($\boldsymbol{A}^t$) of the current stage model for calculation. To enhance the model's memory of old classes, the KD module uses different model information from the current stage($\boldsymbol{A}^t$) and the previous stage($\boldsymbol{A}^{t-1}$).

B. Soft Label Generating

1) For the first stage of training, the labels only include background and old classes. The background pixels received in the first stage may contain new classes that may appear in subsequent stages. Therefore, the weight of the background is reduced accordingly based on the proportion of each class to the total number of classes. The calculation formula is as follows:

$$w_{c_i} = \begin{cases} g\left(\dfrac{\frac{\mathrm{N}}{n_{c_i}}}{\mathrm{len}\,(C^{t'})+1}\right), & c_i \in C^b \\ g\left(\dfrac{\mathrm{N}}{n_{c_i}}\right), & c_i \in C^t \end{cases} \tag{2}$$

where g normalizes the output to the interval 0–1.

Throughout the training process, the reduction of background weight enables the model to focus more effectively on accurately identifying the relevant classes. This modification is designed to mitigate the risk of future misclassifications of classes as background. As a result, the decrease in background weight improves the model's sensitivity and resilience to novel classes.

If $t > 0$, indicating the incremental phase, the labels for new classes can be determined to be completely correct. Therefore, this part can be calculated using the normal CE function. The remaining part, which includes the background and old classes labels, is considered uncertain.

Thus, the CE is first computed for the part corresponding to new classes. The labels for new classes are the true labels for the current stage and are considered as hard labels. The rest of the image is initially set to 255 and is ignored when computing the first part. The formula is as follows:

$$m_i = \begin{cases} g_i^{t-1}, & p_x^{c_j} \geq \eta_{c_j}, y_i \in C^b, g_i^{t-1} = c_j \in C^{t-1} \\ 255, & p_x^{c_j} < \eta_{c_j}, y_i \in C^b, g_i^{t-1} = c_j \in C^{t-1} \\ y_i, & \text{else.} \end{cases} \tag{3}$$

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The loss function for this part is calculated as follows:

$$l_{\text{certain}} = -\sum_{c_i} \frac{w_{c_i}}{\mathrm{N}} \sum_{j=1}^{\mathrm{N}} \log\left(q_j\right) \times q_j \tag{4}$$

where $c_i \in C^b \cup C^{t-1} \cup C^t$,$q_i$ denotes the probability that the pixel belongs to the most likely category inferred by the current model.

2) The frequency of training stages is on the rise, which has implications for the softmax function as it computes the probabilities for all classes collectively. This collective approach may result in the emergence of new classes that overshadow the predictions of existing ones, thereby diminishing the probabilities associated with the older classes. To mitigate this concern, the sigmoid function is employed, as it computes the probability for each class independently, thereby eliminating the influence of other classes. Nevertheless, due to the comprehensive information retained in the softmax function, it continues to be utilized in the calculation process.

In the second part, when the previous stage of training ends, the model saves the predicted Acc and intersection over union (IoU) values for each land cover class. The output matrix of the model for the image in the previous stage is denoted as $\boldsymbol{A}^{t-1}$. The softmax and sigmoid functions are applied to transform the output matrix, resulting in transformed matrices denoted as $\boldsymbol{B}^{t-1}$ and $\boldsymbol{D}^{t-1}$, respectively. The transformation formula for the sigmoid function is as follows:

$$d_i^{t-1} = \frac{1}{1 + e^{-a_i^{t-1}}} \tag{5}$$

where a_i^{t-1} and d_i^{t-1}, respectively, represent the elements of matrices $\boldsymbol{A}^{t-1}$ and $\boldsymbol{D}^{t-1}$.

Through the transformation, two matrices $\boldsymbol{B}^{t-1}$ and $\boldsymbol{D}^{t-1}$ are obtained. The elements b_i^{t-1} and d_i^{t-1} are found in these matrices. These elements represent two types of probabilities. b_i^{t-1} represents the softmax probability. d_i^{t-1} represents the sigmoid probability. Each probability corresponds to a pixel in an image. Specifically, it indicates the likelihood of the pixel belonging to the most probable class. In addition, the most probable specific class that each pixel belongs to after the transformation by the softmax and sigmoid functions is recorded. These two matrices are denoted as $\boldsymbol{E}^{t-1}$ and $\boldsymbol{F}^{t-1}$. The elements e_i^{t-1} and f_i^{t-1} in these matrices represent the most probable specific class that each pixel in each image belongs to.

Next, the calculation of the soft label values for the old classes is discussed. In this study, only the case where $e_i^{t-1} = f_i^{t-1}$ is considered. Through specific experiments, it has been observed that the occurrence of nonmatching elements in matrices $\boldsymbol{E}^{t-1}$ and $\boldsymbol{F}^{t-1}$ is extremely rare, accounting for less than 0.001% of the total pixel points in the image. Therefore, these pixels can be neglected.

Under the aforementioned assumption, where the softmax function and sigmoid function have consistent labeling for classes, only the specific values need to be discussed.

In this experiment, IoU is used as the threshold for the softmax function, while Acc is used as the threshold for the sigmoid function. Since IoU is influenced by other classes, it corresponds to the softmax function, which is also affected by other classes. Therefore, IoU is used as the threshold for the softmax function. Conversely, the Acc of an individual class is not influenced by other classes, and the sigmoid function only calculates the output for a single class, without including values from other classes. Therefore, Acc is used as the threshold for the sigmoid function.

The formula for the specific value calculation is as follows:

$$m_i = \begin{cases} \dfrac{1}{3} + \dfrac{b_i^{t-1}}{3} + \dfrac{d_i^{t-1}}{3}, & b_i^{t-1} >= u_1, d_i^{t-1} >= u_2 \\ \dfrac{e}{\text{sum}} + \dfrac{e^{b_i^{t-1}2}}{\text{sum}} + \dfrac{e^{d_i^{t-1}2}}{\text{sum}}, & b_i^{t-1} >= u_1, d_i^{t-1} < u_2 \\ \dfrac{e}{\text{sum}} + \dfrac{e^{b_i^{t-1}2}}{\text{sum}} + \dfrac{e^{d_i^{t-1}2}}{\text{sum}}, & b_i^{t-1} < u_1, d_i^{t-1} >= u_2 \\ 255, & b_i^{t-1} < u_1, d_i^{t-1} < u_2 \end{cases} \tag{6}$$

where sum is calculated using the following formula:

$$\text{sum} = e + e^{e_i^{t-1}} + e^{f_i^{t-1}} \tag{7}$$

the above conditions hold true only under the condition that $m_i = y_i, g_i^{t-1} = c_j \in C^{t-1}$ in the formula.

The weighted coefficients are denoted as w_a, w_b, and w_c. The assignment formula for soft labels is as follows:

$$m_i = w_a \times 1 + w_b \times b_i^{t-1} + w_c \times d_i^{t-1}. \tag{8}$$

In the most extreme case, since there are three parameters, if both b_i^{t-1} and d_i^{t-1} are equal to 1, the final result will always be 1. The calculation of assigning soft labels essentially involves weighting the hard label 1, the softmax predicted probability b_i^{t-1}, and the sigmoid predicted probability d_i^{t-1}.

So, $w_a = w_b = w_c = (1/3)$, $m_i = 1 \times (1/3) + b_i^{t-1} \times (1/3) + d_i^{t-1} \times (1/3) = 1$, in this case, the soft label is consistent with the hard label.

In general, both b_i^{t-1} and d_i^{t-1} are less than 1. It is also necessary to ensure that the final value of m_i falls within the range of [0, 1].

In the first case, the pixel belongs to the old class. Both the softmax and sigmoid functions predict the class with confidence above the set threshold. Under these conditions, the following result can be obtained:

$$m_i = \frac{1}{3} \times 1 + \frac{1}{3} \times b_i^{t-1} + \frac{1}{3} \times d_i^{t-1} = \frac{1 + b_i^{t-1} + d_i^{t-1}}{3} \tag{9}$$

$$b_i^{t-1} + b_i^{t-1} + b_i^{t-1} < 1 + b_i^{t-1} + d_i^{t-1}. \tag{10}$$

Therefore, after the transformation, the values are higher than the probabilities calculated by the original softmax function. This increases the values of the soft labels and reduces the phenomenon of class interference.

In the case, where either b_i^{t-1} or d_i^{t-1} is below the threshold, it indicates that the model is not completely certain about whether the pixel belongs to the old class. In other words, the model's confidence in predicting this pixel is lower than in the first case. In this situation, for the three weights, a softmax function transformation can be applied to the weights, followed by weighting.

In the last case, if both b_i^{t-1} and d_i^{t-1} are below the threshold, it indicates that the model lacks confidence in predicting the

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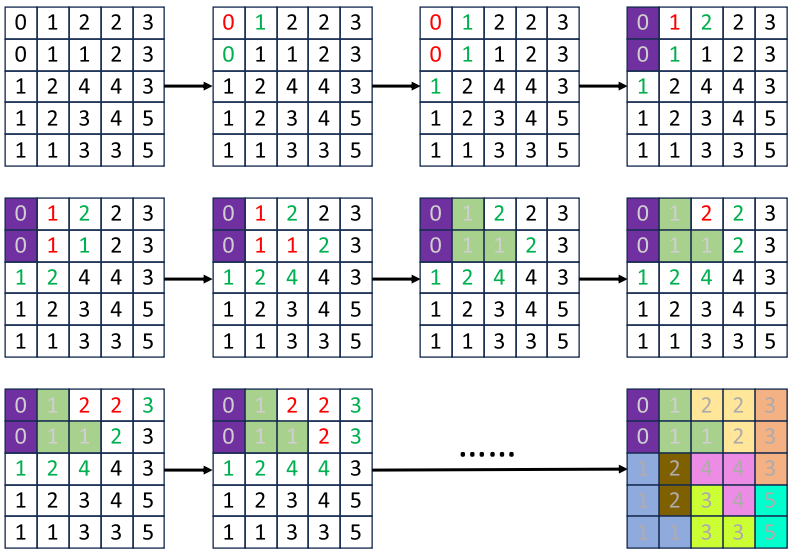


Fig. 2. Illustration of the process of generating subregions.

class for that pixel. Therefore, the pixel is hidden and not included in the calculations.

The previous stage of the model could not accurately predict specific categories. Pseudo labels may not be generated correctly in some cases. This happens when the threshold value of certain pixel points is lower than the calculated threshold value. To address this, this chapter adopts a two-part strategy. First, it uses pixel values that are confident enough for the previous part of the calculation. Second, it also incorporates pixels with a probability less than the threshold value into this part. The risk of incorrect labeling is reduced by including the probability of each class in the computation. The calculation formula is shown below

$$l_{\text{uncertain}} = -\frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \hat{a}_i^t \cdot a_i^{t-1} \tag{11}$$

where $\hat{a}_i^t$, a_i^{t-1} represent the elements of $\hat{A}^t$ and A^{t-1}, respectively, where $\hat{A}^t$ is the model output matrix of the current stage, excluding the new category part, and $\hat{n}$ represents the number of pixels participating in 11.

The condition is that the current label is C^b, the pseudo-label is C^{t-1}, and the probability of belonging to the pseudo-label is less than a threshold, the labels of the pixels that satisfy the requirement are all indeterminate old categories.

The overall CE function is defined as follows:

$$\text{loss}_{\text{ce}} = l_{\text{certain}} + \gamma \cdot l_{\text{uncertain}} \tag{12}$$

where γ is an adaptively adjusted parameter that controls the proportion of new and old classes in the CE calculation. It can be automatically adjusted based on different input images. The specific calculation formula is as follows:

$$\gamma = \frac{N - n_t}{n_t} \tag{13}$$

where N represents the total number of pixels in the image and n_t represents the total number of pixels belonging to the new class. The numerator represents the sum of pixels in the image belonging to the background and old classes. Depending on the situation, γ adaptively adjusts the weights of the two parts in the CE, balancing the proportion of the new and old classes in a reasonable manner, thus guiding network training.

C. Subregion Distillation

1) Generation of Subregions: The BFS algorithm is commonly used for traversing and searching graphs. Subregions can be generated using BFS by exploring adjacent regions in the graph and gradually layering them. To generate new subregions, we search and expand regions in a graph based on given conditions or rules. Breadth-first search expands layer by layer, continuously adding neighboring regions that satisfy the generation condition to the queue and marking them as visited. This allows for the gradual generation of contiguous subregions while ensuring their adjacency. The process of generating subregions using breadth-first search is illustrated in Fig. 2.

Compared to regular optical images, remote sensing images typically contain more subregions with similar features. To better address these challenges, we propose a novel KD method optimized for remote sensing images. First, the proposed method generates subregions. Then, it calculates the similarity between the new and old models for each subregion. Finally, it aggregates these similarities using weighted summation. In actual remote sensing images, the distribution of objects is often irregular, forming unique shapes for rivers, lakes, and grasslands. To handle this irregularity, we consider distillation based on actual irregular subregions, which is a more scientific and reasonable approach. Fig. 3 illustrates the remote sensing image and the resulting subregions, where different colors represent different subregions, which are the parts to be distilled in this experiment.

To address this issue, we adopt a breadth-first search approach using pseudo-labels. Specifically, the new classes in the input image are first identified. Then, using the previously frozen old model, pseudo-labels containing the old classes are

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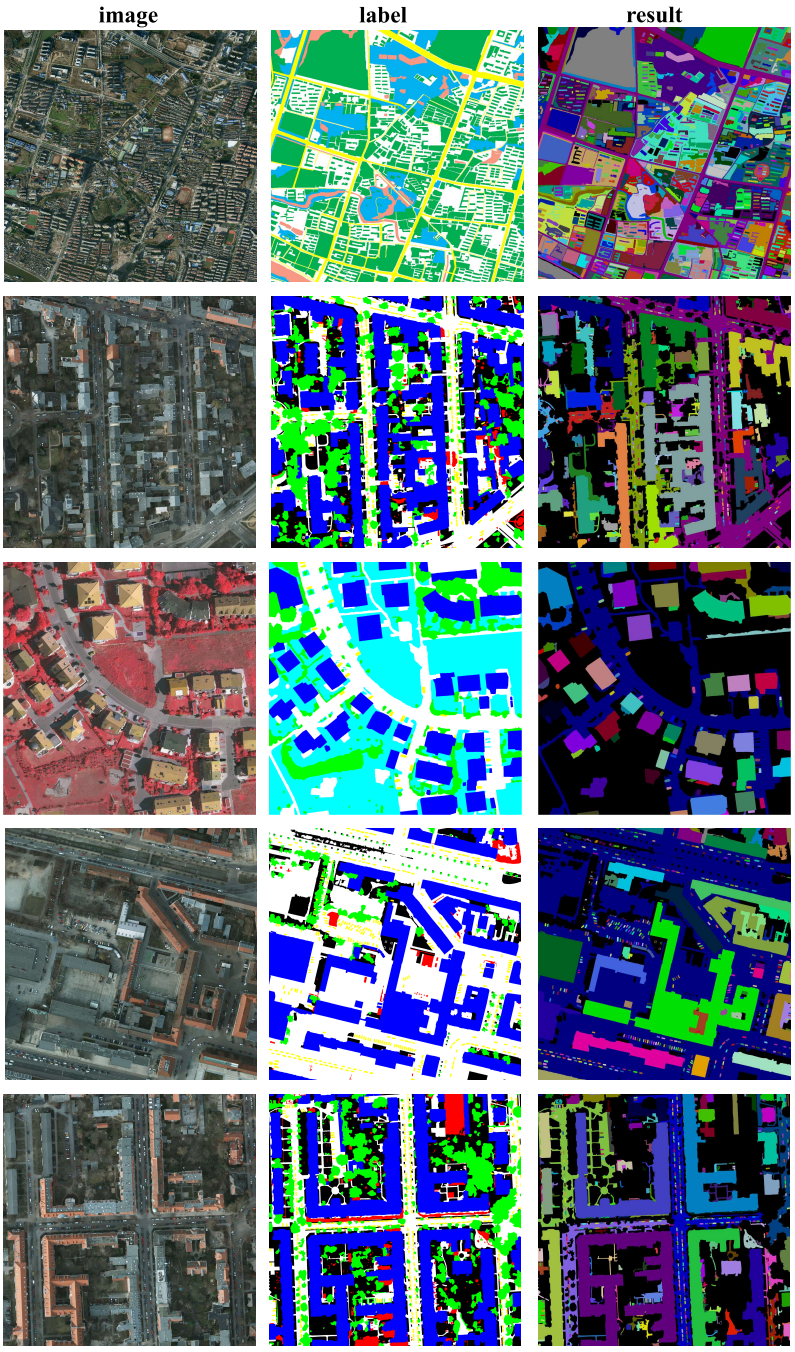


Fig. 3. Remote sensing image and resulting subregion illustration.

generated for the input image. Subsequently, by overlaying the old classes onto the parts of the image excluding the new classes, labels containing both old and new classes are obtained. These labels can be used for the breadth-first search process to generate subregions. When generating subregions, each subregion should be labeled with a unique value. Only the parts corresponding to the old classes are processed during subregion generation. The parts corresponding to the new classes remain unchanged. This is because the old and new models learn different features for the new class parts, and thus, there is no need for distillation in these regions. To avoid confusion and overlapping in the generated regions, it is necessary to start the labeling of subregions from a value higher than the total number of classes in the dataset. The subregion graphics generated based on the remote sensing datasets are shown in Fig. 3. And the process of SRD is illustrated in Fig. 4.

2) Computing Weights: In image recognition tasks, there can be variations in the difficulty of recognizing different subregions. In traditional methods, all subregions are typically assigned the same weight, which may result in some important subregions not receiving sufficient attention. To address this issue, we propose a design for adaptive weights for subregions.

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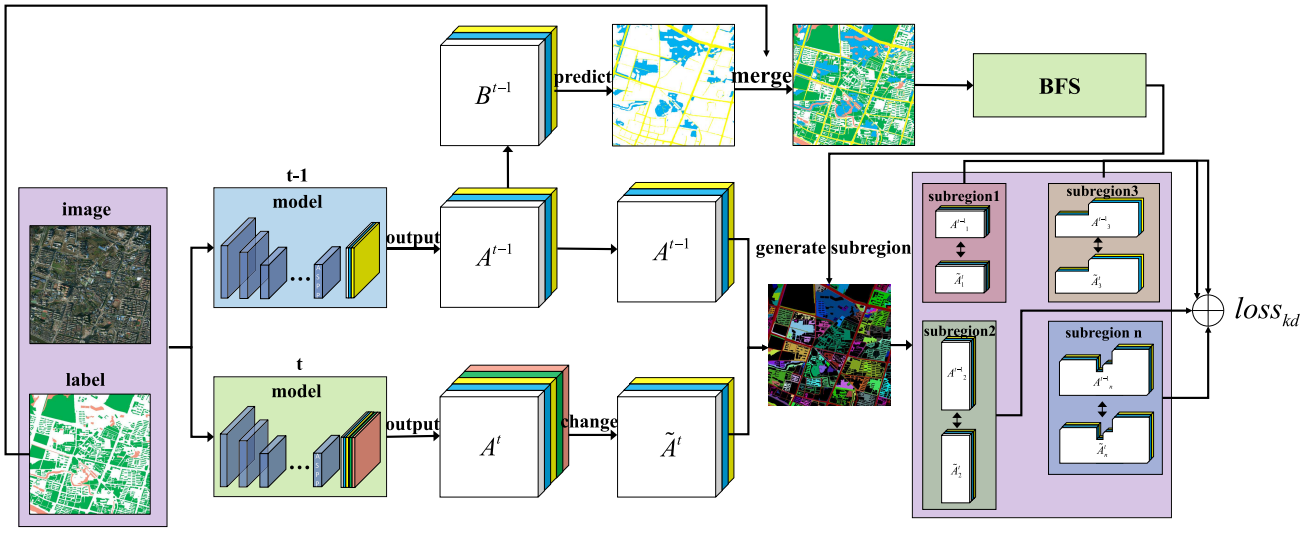


Fig. 4. Distillation process illustration.

The confidence level of the model for individual pixels can be characterized by the softmax output generated by the new model for a given image. Given that the complexity of recognizing various subregions differs, the model's confidence levels across these subregions will also vary. By modulating the model's confidence levels for distinct subregions, the efficacy of the training process can be enhanced.

To ensure the effectiveness of model training, this experiment adjusts the weights based on the model's confidence level for different subregions. To achieve this, the confidence level of the model for individual pixels was defined as the softmax output of the model for the image, reflecting the model's confidence in its predictions for each pixel. Subsequently, adaptive weights for subregions were designed based on this characteristic.

Specifically, we calculate the confidence level for each subregion and uses it as the weight for that subregion. This design of adaptive subregion weights helps improve the training effectiveness of the model. It ensures that the model pays more attention to important subregions, thereby improving recognition Acc and performance. Moreover, this weight adjustment method can adapt to the characteristics of different images, further enhancing the model's generalization ability.

During the distillation process, the background and parts of the old classes in the pseudo-labels are analyzed. These regions often contain incorrect information. This incorrect information originates from the inference of the old model. Therefore, we measure the model's confidence level for subregions by calculating the softmax probabilities of the model for each pixel in the subregion. Then, the weights of the subregions are adjusted based on these confidence levels.

Through this design, based on the confidence levels of the subregions, the weights can be adjusted more accurately to ensure the effectiveness and fairness of the distillation process. This approach takes into account both the model's confidence level for different subregions and the number of pixels in each subregion, providing a better reflection of the importance of the subregions.

The formula for measuring the confidence level of a subregion is as follows:

$$\text{certain}_{\text{sub}_i} = \sum_{j=1}^{N} -p * \log(p). \tag{14}$$

By iterating through each subregion, we can obtain the confidence level of each subregion, which allows us to calculate the overall confidence level of all subregions (the distilled part of the image). The formulas for the calculations are as follows:

$$\text{total}_{\text{certain}} = \sum_{i=1}^{N} \text{certain}_{\text{sub}_i} \tag{15}$$

for the weight of subregion i, the calculation formula is

$$w_{\text{sub}_i} = \frac{\text{total}_{\text{certain}} \cdot \sqrt{\text{cnt}_{\text{sub}_i}}}{\text{certain}_{\text{sub}_i}} \tag{16}$$

where $\text{cnt}_{\text{sub}_i}$ represents the number of pixels contained in subregion i, and it is incorporated into the weight calculation by taking its square root. This value can be maintained and updated during the iteration of subregions, allowing for constant-time access $O(1)$ to this value in the subsequent calculations.

The distillation loss function can be calculated once the subregional weights have been determined.

The specific formula for the calculation is as follows:

$$\text{loss}_{\text{kd}} = \sum_{i=1}^{N} w_{\text{sub}_i} \cdot \sum_{j=1}^{M} \log\left(f\left(a_i^{t-1}\right)\right) \cdot f\left(\tilde{a}_i^{t}\right). \tag{17}$$

D. Overall Loss Function

The overall loss function in this chapter gradually increases the regularization coefficient in the distillation function. It can be defined as follows:

$$\text{loss}_{\text{total}} = \alpha \cdot \text{loss}_{\text{ce}} + \beta \cdot f\left(\frac{e_t \cdot s_t}{e_n \cdot s_n}\right) \cdot \text{loss}_{\text{kd}} \tag{18}$$

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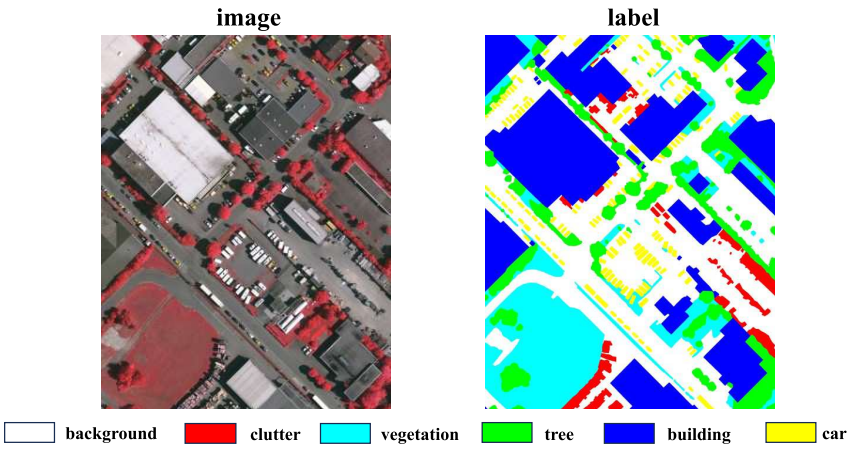


Fig. 5. Images and labels of the Vaihingen dataset.

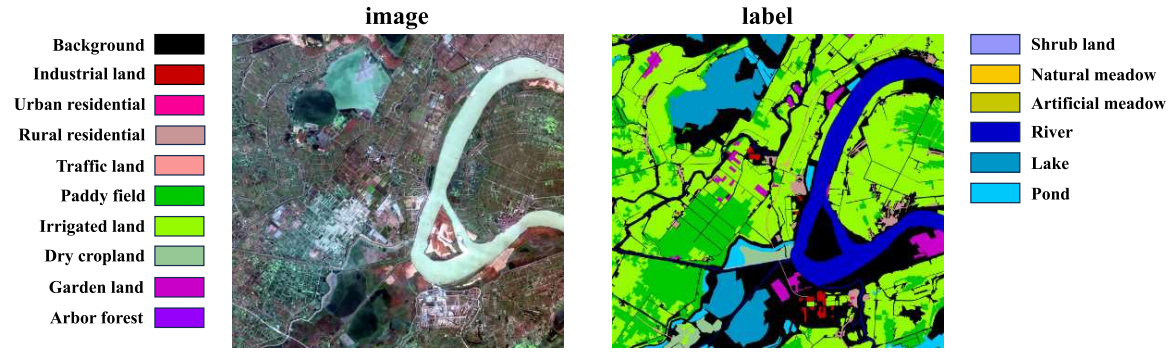


Fig. 6. Images and labels of the GID dataset.

where α and β are hyperparameters that can be adjusted to balance the different loss components.

In this equation, loss_{ce} represents the CE loss, loss_{kd} represents the distillation loss, and $f((e_t \cdot s_t/e_n \cdot s_n))$ represents a function that gradually increases the regularization coefficient. The terms e_t, s_t, e_n, and s_n are the logits or probabilities from the teacher and student models, respectively.

It is important to adjust the values of α and β to achieve the desired balance between the CE loss and the distillation loss in the overall loss function.

IV. Experimental Analysis and Results

A. Experiment Preparation

1) Dataset: Dataset Vaihingen is provided by ISPRS [53]. There are many detached buildings and small multistory buildings in Vaihingen, which is a relatively small village. In the dataset, we use images of RGB bands. The spatial resolution is 9 cm. The Vaihingen dataset has been classified manually into six categories: background, building, car, tree, vegetation, and clutter. The partial images and labels of the Vaihingen dataset are shown in Fig. 5.

The GID dataset is a large-scale, high-resolution land cover dataset constructed using data from China's Gaofen-2 satellite. It consists of two parts: the large-scale classification set (GID-5) and the fine land cover set (GID-15). The dataset we used is GID-15.

The GID-15 dataset includes 30 000 image patches and covers 15 land cover categories, including paddy fields, irrigated land, dryland, orchards, deciduous forests, shrub forests, natural grasslands, artificial grasslands, industrial land, urban residential areas, rural residential areas, transportation land, rivers, lakes, and ponds. The partial images and labels of the GID dataset are shown in Fig. 6.

The FBP dataset contains 5 billion pixels. It has a spatial resolution of 4 m and covers an area of over 50 000 km^2 in China. The dataset includes over 5 billion annotated pixels. It has the advantages of rich category diversity, large coverage, wide distribution, and high spatial resolution. These characteristics enable it to effectively reflect the distribution of real-world land features, making it beneficial for various land cover-related research studies. The partial images and labels of the FBP dataset are shown in Fig. 7.

These datasets are large enough and representative of real-world remote sensing scenarios. They contain diverse land cover types, ensuring a comprehensive assessment of the proposed method.

2) Implementation Details: We follow the parameters set in MiB for all methods. Resnet101 is used as the backbone and deeplabv3 is used as the model. A pretrained model of ImageNet has been used to initialize the backbone. We use the SGD algorithm for the gradient descent, with a corresponding decay of the learning rate over training time, with a decay rate of 0.9. In the incremental stage setting, the initial learning

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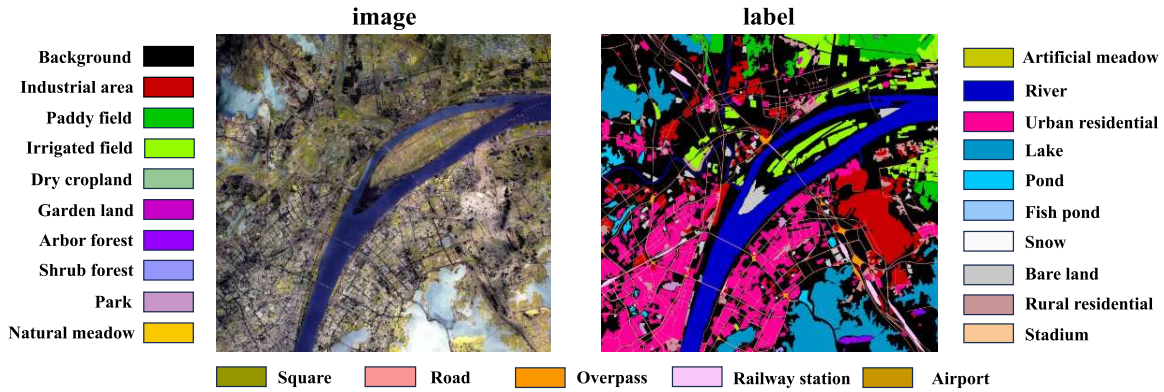


Fig. 7. Images and labels of the FBP dataset.

rate is 10^{-2} for the first stage, and 10^{-3} for the subsequent stages. α is set at 1 and β is set at 30. Training epochs for the Potsdam and Vaihingen datasets are 10 and 20 for the CCF dataset, respectively. During the first stage of training, we use a batch size of 12. During the next incremental stage, we trained the model with a batch size of 10. In the Vaihingen dataset, the first 25 images are selected as the training set and the remaining 8 images are selected as the validation set.

In the traditional method, for an input tensor of dimensions [12, 3, 512, 512], hundreds of subregions need to be processed, which means hundreds of iterations are required to complete the calculation. Even though PyTorch optimizes for loops, this approach is still inefficient when processing large remote sensing images. In terms of complexity, the time complexity of the traditional method is $O(bs \times W \times H)$. In contrast, our optimized method completes the distillation in one computation through matrix operations, greatly improving efficiency. Specifically, we first compute the probability matrix for all pixel points, then apply the log transformation and matrix multiplication to obtain the confidence level for each pixel. This avoids pixel-by-pixel traversal and accelerates the computation process. We also use the torch.bincount function to efficiently calculate the confidence levels of subregions, which further optimizes the entire process. Therefore, our method significantly reduces both time and memory complexity.

B. Comparative Methods

We choose FT [54], EWC [40], RW [41], PI [39], ILT [42], LWF [14], LWF-MC [15], MiB [43], PLOP [44], LSAW [55], and SLSRD as the comparison algorithms for our experiments. We compare them with our proposed algorithm. Joint represents that the model is trained on all classes, and it represents the upper limit of incremental learning performance. Table I represents the results of the algorithm on the Vaihingen datasets. Due to the large number of classes in the GID and FBP datasets, we have set more incremental tasks for these two datasets to explore the performance of different algorithms. Tables II and III represent the results of algorithms on the GID dataset under different incremental tasks. Tables IV –VI represent the results of algorithms on the FBP dataset under different incremental tasks. Figs. 6–8 represent the predicted results of algorithms on the Vaihingen, GID, and FBP datasets. The evaluation metrics include mean Acc (MA) and mean IoU (mIoU). Our method achieves the best results.

The GID and FBP datasets serve as good extensions to the Vaihingen dataset. Considering the increasing number of land cover categories, more incremental stages can be set to better simulate the continuous growth of land cover categories in real life. This allows for a more comprehensive evaluation of incremental land cover classification methods and better adaptation to potential new classes that may emerge in the future.

In the presentation of the results of the GID and FBP datasets, due to the large number of categories, only two incremental tasks with the highest number of stages (3-3-3-3-3 and 6-3-3-3-3-3-3) are selected for demonstration. For other incremental tasks, the mIoU of all newly learned classes in each stage is used to represent the learning results for that stage. The notation (3-3-3-3-3) denotes an incremental learning task configuration, where each number represents the number of new categories (excluding background categories) introduced in each stage. For example, (3-3-3-3-3) means that three new categories were added in each of the five incremental stages. Next, an analysis will be conducted on the specific tasks of each dataset.

1) Quantitative Analysis on the Vaihingen Dataset: a) (3-2): Two learning steps were performed in this experiment. Four feature categories, including background, were learned for both datasets in the first phase, and the remaining two categories were learned in the second phase. For the Vaihingen dataset, the FT, EWC, RW, and PI algorithms all have serious forgetting problems for the car category, while ILT, LWF, LWF-MC, MiB, PLOP, and LSAW methods still maintain good results. Due to the small number of categories that clutter occupies in an image, all algorithms are less effective at learning it. The SLSRD algorithm achieves the highest mIoU on the Vaihingen dataset of 68.9%, which is a 7.5% improvement over the MiB algorithm, while retaining the ability to memorize the old categories. It also reached the highest IoU on the background class, 84.3% on the Vaihingen dataset, which fully demonstrates that the SLSRD algorithm memorizes the old feature elements well while learning new feature classes, and at the same time effectively mitigates the background bias problem.

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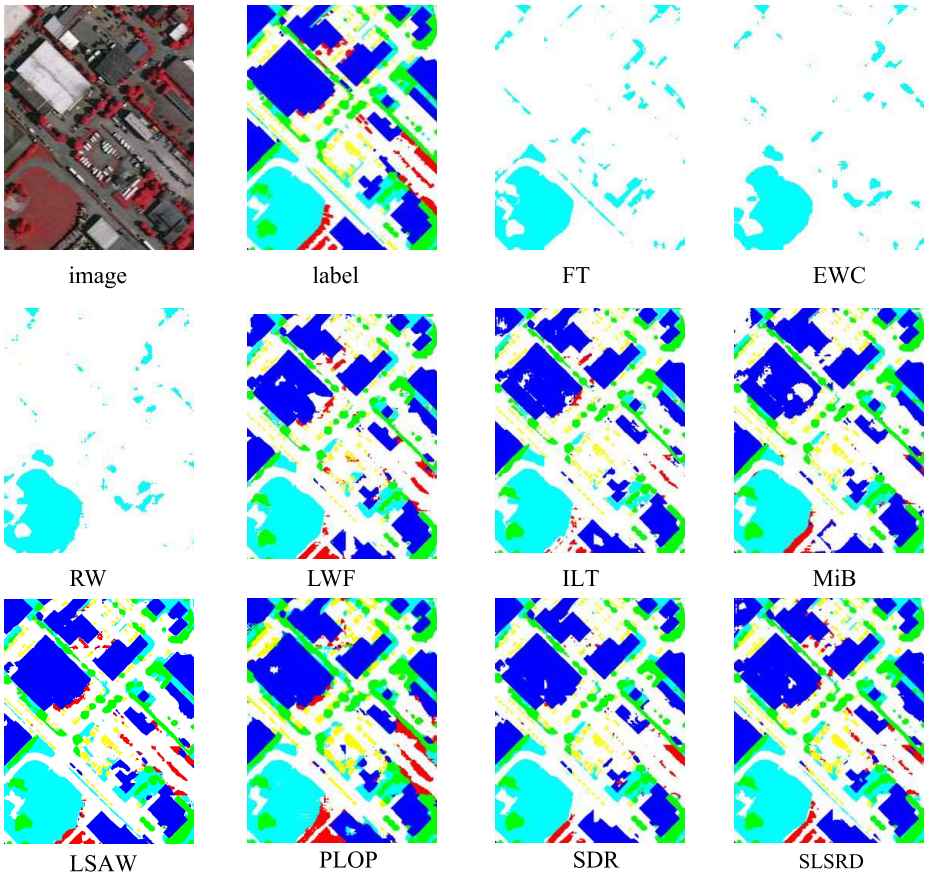


Fig. 8. Experimental results on the Vaihingen dataset.

b) (3-1-1): Three learning steps were performed in this experiment. Four feature categories, including background, were learned for both datasets in the first stage, and the remaining two categories were learned sequentially in the second and third stages. In the last step of this experiment, the small number of clutter feature categories posed a challenge for the algorithms to maintain the ability to memorize this category, and most algorithms did not memorize this category as well. For the Vaihingen dataset, most of the compared algorithms had severe catastrophic forgetting problems on the clutter category, where MiB's recognition IoU for the clutter category decreased from 18.8% to 11.9% for the previous task, and PLOP's recognition IoU for the clutter category decreased from 20.3% to 0.3% for the previous task, but the LWF algorithm maintained its memory for the clutter category, which only decreased from 22.1% to 21.9%, and was not affected by this task, and finally achieved 60.0% mIoU. The LSAW algorithm also maintained its memory for the clutter category, and was not affected by this task either, reached 64.6% mIoU. The SLSRD algorithm in this article still achieved the highest IoU of 37.5% for the clutter category, and 66.2% mIoU for the overall category, which is a 1.6% improvement over the best-performing LSAW algorithm.

c) (2-2-1): Three learning steps were performed in this experiment. Three feature categories, including background, were learned for both datasets in the first stage, the fourth and fifth categories were learned in the second stage, and the sixth category was learned in the third stage. It should be noted that the model initially acquired less information about the data due to one less category in the first stage, which made all algorithms less effective than the results of the previous experiments. For the FT, EWC, RW, and PI algorithms, they are all effective on the last category, and observing the IoU and Acc data at the same time, it can be seen that the above algorithms have an IoU of about 40% on the background category, but the Acc of the background category reaches more than 97%. This shows that the above algorithms recognize other learned old classes as background classes, which creates a serious background bias problem. Most of the other compared algorithms improved their memory for clutter classes compared to the 3-1-1 task due to the differentiation produced by learning the clutter classes at the same time the model learns the tree classes, which also illustrates that the incremental task with a smaller sample size and the inclusion of only one feature class in a single stage is more challenging. The SLSRD algorithm achieved 68.2% of the mIoU value on the Vaihingen dataset, which is higher than the 3-1-1 task of the mIoU value, which is 2.2% higher than the best performing LSAW algorithm.

2) Quantitative Analysis on the GID Dataset: a) (10-5): In this experiment, two learning steps are performed. The first stage of learning includes 11 classes, including the background, and the second stage of learning involves the remaining five classes. For the FT algorithm, as it only fine-

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TABLE I
EXPERIMENTAL RESULTS ON THE VAIHINGEN DATASET

Algorithm	background	building	car	tree	clutter	vegetation	mIoU
				3-2			
FT	47.4	0	0	0	25.5	57.6	21.7
EWC	45.9	0	0	0	10.6	59.9	19.4
RW	45.5	0.2	0.1	0	12.4	59.7	19.7
PI	47.4	0	0	0	25.4	57.8	21.7
ILT	77.3	85.7	56.0	61.9	8.0	58.5	57.9
LWF	79.3	85.8	55.7	61.3	22.1	60.2	60.7
MiB	81.1	84.6	55.4	65.9	18.8	62.8	61.4
PLOP	81.1	84.5	58.1	52.1	20.3	55.8	58.6
SDR	82.4	88.0	58.3	**67.8**	27.5	63.4	64.6
LSAW	83.8	87.1	61.4	63.6	33.4	63.9	65.5
SLSRD	**84.3**	**88.7**	**66.7**	67.0	**40.6**	**65.8**	**68.9**
				3-1-1			
FT	46.5	0	0	0	0	61.7	18.0
EWC	44.7	0	0	0	0	58.8	17.2
RW	44.2	0	0	0	0.2	55.8	16.7
PI	46.5	0	0	0	0	61.8	18.0
ILT	73.9	86.4	55.0	62.8	19.7	54.7	58.7
LWF	76.2	86.5	53.9	62.3	21.9	59.3	60.0
MiB	79.4	84.7	54.7	**68.2**	11.9	59.1	59.7
PLOP	79.3	84.9	57.7	53.2	0.3	58.3	55.6
SDR	76.8	79.4	53.7	64.0	28.3	56.9	59.9
LSAW	79.5	**87.3**	61.7	61.6	33.9	63.7	64.6
SLSRD	**80.9**	85.5	**63.0**	64.9	**37.5**	**65.6**	**66.2**
				2-2-1			
FT	46.4	0	0	0	0	63.5	18.3
EWC	43.9	0	0	0	0	51.8	15.9
RW	42.8	0	0	0	0	35.1	13.0
PI	46.4	0	0	0	0	63.4	18.3
ILT	70.4	86.2	56.3	51.2	0.1	47.0	51.9
LWF	75.8	86.8	56.7	60.6	10.1	56.9	57.8
MiB	74.0	84.9	56.0	60.2	25.6	27.3	54.7
PLOP	73.6	84.3	50.7	59.1	28.0	50.5	57.7
SDR	81.1	87.7	59.3	65.5	36.8	60.5	65.2
LSAW	80.9	86.3	63.3	63.7	**38.0**	63.9	66.0
SLSRD	**85.5**	**89.3**	**66.3**	**67.8**	34.1	**66.2**	**68.2**

TABLE II
EXPERIMENTAL RESULTS ON THE GID DATASET

Algorithm	10-5			6-6-3				3-3-3-3-3		
	0-10	11-15	mIoU	0-6	7-13	13-15	mIoU	0-9	10-15	mIoU
FT	3.71	14.24	7.00	5.87	0	8.95	4.24	4.10	3.84	4.00
EWC	3.83	16.27	7.72	5.86	0.44	8.51	4.33	4.13	6.72	5.10
RW	3.82	13.85	6.95	6.05	1.24	10.8	5.14	4.25	16.50	3.68
PI	3.72	13.18	7.06	5.85	0	8.55	4.16	4.08	3.96	4.03
ILT	33.32	23.11	30.13	42.08	15.91	22.11	28.52	24.06	**20.14**	22.59
LWF	33.21	19.16	28.82	41.39	17.48	18.95	28.22	27.30	10.56	21.02
LWF-MC	35.31	10.36	27.52	41.15	3.61	0.89	19.52	15.77	0.79	10.15
MiB	37.49	17.04	31.10	49.07	17.93	7.52	29.60	36.60	11.06	26.33
PLOP	35.74	28.17	33.37	47.47	16.59	**27.44**	32.14	26.78	15.86	21.69
SDR	39.33	26.64	35.37	52.40	**19.27**	12.08	32.42	39.65	12.63	28.72
LSAW	38.24	27.53	34.67	51.66	18.72	18.49	30.18	**40.12**	12.58	29.26
SLSRD	**40.42**	**30.53**	**37.33**	**53.87**	18.90	13.34	**33.16**	37.5	19.51	**30.75**

tunes the previous model without the KD part, it forgets almost all the previous classes, resulting in an mIoU of only 3.71% for the old classes. Other comparative algorithms can remember the old classes well, and the ultimate impact on mIoU is the learning of the new classes. The MiB algorithm hardly learns any new classes, resulting in a lower mIoU. The LWF and LWF-MC algorithms achieve a slightly higher mIoU than the MiB algorithm as they learn some new classes. The PLOP algorithm not only maintains good memory for the old classes but also effectively learns the features of the new

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TABLE III
EXPERIMENTAL RESULTS ON THE GID DATASET (3-3-3-3-3) (FIVE INCREMENTAL STAGES)

Algorithm	FT	ILT	LWF	LWF-MC	MiB	PLOP	SDR	LSAW	SLSRD
Background	41.04	50.67	50.97	42.58	55.14	51.6	57.98	**60.22**	59.75
Industrial land	0	37.92	31.34	40.22	50.79	47.67	54.51	55.24	**57.26**
Urban residential	0	49.69	42.29	47.21	55.91	45.9	57.27	57.82	**61.15**
Rural residential	0	22.76	17.08	27.71	31.23	23.0	28.27	29.16	**31.06**
Traffic land	0	10.52	19.85	0	34.67	13.16	40.47	**46.47**	41.55
Paddy field	0	0.2	9.91	0	27.84	4.59	**45.84**	17.94	9.13
Irrigated land	0	33.65	51.49	0	61.18	51.39	61.66	53.11	**61.74**
Dry cropland	0	0	0	0	**2.52**	0	0.98	0.33	0
Garden land	0	0	0	0	0	0	**0.88**	0.62	0
Arbor forest	0	35.21	50.08	0	46.72	30.46	48.64	51.51	**53.36**
Shrub land	0	0	0	0	0	0	0	0	0
Natural meadow	0	32.35	31.99	0.15	38.26	10.05	43.11	51.41	**54.09**
Artificial meadow	0	0	0.34	0	**0.25**	0	0	0	0
River	6.04	**73.01**	14.85	2.09	2.39	62.09	3.78	53.08	42.59
Lake	10.03	13.56	12.32	2.49	7.27	5.86	10.04	15.12	**17.41**
Pond	6.97	1.96	3.83	0.01	**7.14**	1.3	6.24	3.861	3.0
mIoU	4.0	22.59	21.02	10.15	26.33	21.69	28.73	29.26	**30.75**

TABLE IV
EXPERIMENTAL RESULTS ON THE FBP DATASET

	19-5			15-9			16-8		
Algorithm	0-19	20-24	mIoU	0-15	16-24	mIoU	0-16	17-24	mIoU
FT	1.51	23.67	5.94	1.9	25.0	10.21	1.79	24.97	9.21
EWC	1.54	14.41	4.11	1.97	16.27	7.12	1.86	18.19	7.09
RW	1.54	13.31	3.89	1.9	16.49	7.16	1.79	13.72	5.61
PI	1.51	21.55	5.52	1.9	21.78	9.06	1.79	24.55	9.07
ILT	42.1	7.76	35.23	49.1	12.82	36.04	42.74	13.89	33.51
LWF	40.96	11.19	35.01	46.84	11.75	34.21	43.36	16.51	34.77
LWF-MC	**44.79**	3.47	36.53	49.7	6.84	34.27	43.97	6.82	32.08
MiB	43.05	7.93	36.03	50.13	15.02	37.49	47.43	16.34	37.48
PLOP	43.56	7.55	36.36	49.08	12.61	35.95	45.04	13.86	35.06
SDR	44.73	16.06	39.0	**51.01**	19.14	39.53	46.96	20.49	38.49
LSAW	44.23	29.56	41.69	45.26	22.27	41.94	47.81	22.21	42.54
SLSRD	44.3	**36.61**	**42.76**	48.46	**36.19**	**44.05**	**47.92**	**35.86**	**44.06**

TABLE V
EXPERIMENTAL RESULTS ON THE FBP DATASET

	8-8-8				12-3-3-3-3					
Algorithm	0-8	9-16	17-24	mIoU	0-12	13-15	16-18	19-21	22-24	mIoU
FT	3.39	0	**39.87**	13.98	2.34	0	30.09	15.46	29.44	10.21
EWC	3.36	0	19.91	7.58	2.42	0	26.42	12.26	10.14	7.12
RW	3.49	0	17.66	6.91	2.34	0.01	26.9	12.07	10.51	7.16
PI	3.37	0	28.33	10.28	2.34	0	30.18	15.44	19.72	9.06
ILT	43.0	37.22	13.07	31.57	48.16	53.2	26.38	12.08	0	36.04
LWF	45.2	41.53	21.0	36.28	45.83	51.23	20.39	9.16	5.7	34.21
LWF-MC	44.49	6.98	5.65	20.06	48.63	54.31	16.38	4.13	0	34.27
MiB	48.18	44.47	16.2	36.76	49.19	54.22	30.46	14.6	0	37.49
PLOP	44.11	37.43	11.88	31.66	47.9	54.18	25.49	12.33	0	35.95
SDR	49.59	43.91	20.43	38.44	**50.22**	54.43	28.75	14.38	14.27	39.53
LSAW	46.24	45.23	31.19	41.89	49.98	44.85	29.26	18.71	19.56	41.24
SLSRD	**49.41**	**48.28**	36.60	**44.95**	**49.95**	**55.24**	**31.50**	**20.81**	**37.51**	**43.38**

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TABLE VI
EXPERIMENTAL RESULTS ON THE FBP DATASET (6-3-3-3-3-3-3) (SEVEN INCREMENTAL STAGES)

Algorithm	FT	ILT	LWF	LWF-MC	MiB	PLOP	SDR	LSAW	SLSRD
Background	29.99	35.23	36.01	34.85	51.11	48.59	56.3	47.85	**58.98**
Industrial area	0	1.09	0.78	16.13	**55.01**	53.53	53.3	42.43	52.78
Paddy field	0	1.9	0.45	9.85	56.16	57.58	61.9	41.76	**59.22**
Irrigated field	0	2.13	2.01	51.03	71.61	70.92	**77.43**	56.84	76.18
Dry cropland	0	8.36	0.42	20.5	54.9	44.79	**63.9**	61.49	60.94
Garden land	0	1.73	2.26	2.72	9.35	14.53	**18.25**	16.62	10.33
Arbor forest	0	4.89	8.42	56.11	69.48	66.42	70.13	73.49	**73.9**
Shrub forest	0	0	0	0	8.68	0.59	10.35	6.23	**15.59**
Park	0	0	0	0	0	0	0	12.47	**14.62**
Natural meadow	0	8.94	0.66	0	41.93	40.77	50.4	47.38	**53.88**
Artificial meadow	0	0	7.11	0	13.33	0	14.2	24.96	**19.29**
River	0	4.92	25.25	0	56.04	4.55	46.93	17.44	**60.94**
Urban residential	0	21.99	21.02	0	63.89	59.96	65.87	62.13	67.8
Lake	0	59.16	73.76	0.13	76.2	64.79	55.54	63.34	**79.46**
Pond	0	2.46	9.13	0	24.72	0.04	19.97	24.83	**27.57**
Fish pond	0	40.97	45.14	0.02	52.38	51.41	**59.8**	48.74	56.55
Snow	0	0	0	0	0	0	0	0	0
Bare land	0	9.48	12.82	0.02	30.06	15.73	22.06	34.51	**34.67**
Rural residential	0	40.49	44.34	0.08	54.97	44.87	54.85	54.77	**58.92**
Stadium	0	0	0	0	0	0	0	**22.64**	15.96
Square	0	0	0	0	0	0	0	**6.11**	0.37
Road	0	25.54	36.32	4.98	41.41	24.82	38.97	36.62	**52.16**
Overpass	41.67	2.03	38.77	0	0	0	24.99	14.85	**62.07**
Railway station	0	0	0	0	0	0	0	5.29	**24.26**
Airport	0.16	0	0	0	0	0	0	2.37	**27.75**
mIoU	2.87	10.85	14.59	7.86	33.25	26.56	34.61	32.85	**42.57**

classes, resulting in an mIoU of 33.37%. The LSAW algorithm demonstrates a competitive performance, achieving an mIoU of 34.67%, which is lower than the best-performing SLSRD algorithm but higher than the PLOP algorithm. Compared to the aforementioned comparative algorithms, the proposed method in this chapter achieves optimal performance in both remembering the old classes and learning the new classes, resulting in an mIoU of 37.33%, which is approximately 2% higher than the best-performing SDR algorithm.

b) (6-6-3): In this experiment, three learning steps are performed. The first stage of learning includes seven classes, including the background, the second stage involves six classes, and the third stage involves three classes. Similar to the previous analysis, the FT algorithm forgets almost all the previous classes, and due to the increasing number of training stages, its mIoU further decreases to only 4.24%. For the other comparative algorithms, they can remember the old classes well, and the ultimate impact on mIoU is the learning of the new classes. The MiB algorithm hardly learns any new classes, resulting in a lower mIoU. The LWF and LWF-MC algorithms achieve a slightly higher mIoU than the MiB algorithm as they learn some new classes. The PLOP and SDR algorithms not only maintain good memory for the old classes but also effectively learn the features of the new classes, resulting in an mIoU of 32.14% and 32.42%, respectively. The LSAW algorithm shows robust performance with an mIoU of 30.18%, which is competitive among the compared algorithms. This suggests that LSAW has a reliable capability to handle incremental learning tasks across multiple stages, maintaining a balance between forgetting less and learning more effectively than some of its counterparts. Compared to the aforementioned comparative algorithms, the proposed method in this chapter achieves optimal performance in both remembering the old classes and learning the new classes, resulting in an mIoU of 33.16%, which is approximately 0.7% higher than the best-performing SDR algorithm.

c) (3-3-3-3-3): In this experiment, there are five learning steps. The first stage of learning includes four classes, including the background, and the remaining four stages involve learning three classes each. Due to the increase in the number of stages to five, it presents a higher challenge to the learning capabilities of the models, and the mIoU of all algorithms decreases compared to the previous incremental tasks. Among them, the PLOP algorithm experiences the most significant decline. It can be observed from the figure that only the ILT and PLOP algorithms can effectively recognize the deep blue class (rivers), as shown in Fig. 9. In addition, from Table III, it can be seen that ILT achieves an IoU of 73.01% on rivers, while PLOP achieves 62.09%, far outperforming other algorithms. Other algorithms tend to confuse rivers with the class of lakes, resulting in incorrect recognition of the dominant rivers in the image. LWF and MiB algorithms mostly recognize rivers as lakes, leading to low recognition rates for the river class, which has a significant impact on the overall mIoU reduction. From Fig. 9, it can also be observed that the LWF-MC algorithm has a severe background shift issue as it recognizes most rivers and irrigated land as the background class. This indicates that during the incremental learning stage, as new classes are continuously added, the algorithm tends to classify them as the background class because the new classes are transferred from the background class labels. Therefore, it fails to effectively learn the features of the new classes,

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resulting in a significant background shift. The LWF-MC algorithm achieves only 42.58% IoU on the background class, which is similar to the FT algorithm that already has inherent background shift issues. The other comparative algorithms do not exhibit severe background shift problems. For the SDR algorithm, it achieves a maximum of 45.84% IoU on the rice paddy class, resulting in an overall mIoU of 28.73%. For the LSAW algorithm, it reaches a maximum of 60.22% IoU on the background, resulting in an overall mIoU of 29.26%. Compared to the aforementioned comparative algorithms, the proposed method in this chapter achieves optimal performance in both remembering the old classes and learning the new classes, with an mIoU of 30.75%, which is approximately 1.49% higher than the best-performing LSAW algorithm.

3) Quantitative Analysis on the FBP Dataset: In the study of the FBP dataset, multiple incremental stages were conducted in this chapter. Specifically, up to seven incremental stages were set, with the first stage learning the background and three new classes, and each subsequent incremental stage learning three new classes. Through this incremental learning process, the proposed method in this chapter ultimately achieves optimal recognition performance. In addition, by analyzing the table data, it can be observed that methods like ILT and LWF still achieve relatively high recognition performance when the number of incremental stages is low. However, as the number of incremental stages increases, the Acc of these methods rapidly declines, and the recognition performance significantly deteriorates. This phenomenon may be due to the cumulative effect of class accumulation during the incremental learning process. As the number of incremental stages increases, the model needs to gradually learn more new classes while maintaining the recognition ability for the already learned classes. This challenge can lead to the occurrence of forgetting and interference phenomena when the model faces a large number of new classes, thus affecting the Acc and performance of recognition. For algorithms like ILT and LWF, their CE and loss function designs are not sufficient to cope with such complex situations, resulting in a significant decline in Acc.

a) (19-5)(15-9)(16-8): For tasks 19-5, 15-9, and 16-8, as they all belong to the two-stage incremental learning tasks, we will analyze them together. For the FT, EWC, PI, and RW algorithms, they all suffer from the problem of forgetting the old classes. However, the LWF, LWF-MC, MiB, PLOP, SDR, and LSAW algorithms are able to maintain good memory of the old classes while learning the new classes, resulting in an mIoU of over 32% for all of them. In the 19-5 task, the proposed SLSRD algorithm achieves an mIoU of 42.76%, which is approximately 3.7% higher than the best-performing SDR algorithm. In the 15-9 task, the SLSRD algorithm achieves an mIoU of 44.06%, which is approximately 1.52% higher than the best-performing LSAW algorithm. In the 16-8 task, the SLSRD achieves the highest mIoU for both old and new classes, reaching 44.06%, which is approximately 2% higher than the best-performing LSAW algorithm.

b) (8-8-8): In this experiment, there are three learning steps. The first stage of learning includes 9 classes, including the background, and the second and third stages both involve learning 8 classes. Among them, the LWF-MC algorithm shows a significant decline, achieving over 32% mIoU in the first three incremental tasks, but dropping to 20.06% in this task, indicating that the LWF-MC algorithm is not suitable for this incremental task. Our algorithm ultimately achieves an mIoU of 44.95%, which is approximately 3.06% higher than the best-performing LSAW algorithm.

c) (12-3-3-3-3): This task consists of five incremental stages. The first stage of learning includes 13 classes, including the background, and each subsequent stage learns three classes. The SLSRD algorithm achieves the highest mIoU in all five incremental stages, with a final mIoU of 43.38%, which is approximately 2.14% higher than the LSAW algorithm.

d) (6-3-3-3-3-3-3): The FBP dataset contains 25 classes, including the background. Therefore, it is possible to set multiple incremental stages, and the more stages there are, the higher the robustness requirement for the algorithms. Thus, to better simulate the continuous flow of data in practical scenarios and evaluate the performance of various algorithms in complex incremental tasks, a seven-stage incremental task was set. In this task, the first stage of learning includes seven classes, including the background, and each subsequent stage learns three new classes. As shown in Table VI, the mIoU of all algorithms decreases due to the large number of incremental stages. Fig. 10 provides a visual representation, illustrating the challenges of learning features for numerous land cover types in remote sensing image data. In the previous Vaihingen dataset, the ILT, LWF, and LWF-MC algorithms outperformed the FT, EWC, and RW algorithms, not only in terms of data but also in the quality of predicted results. However, in the case of the FBP dataset with a large number of land cover types and incremental stages, these algorithms still suffer from significant catastrophic forgetting of previously learned old classes. In addition, it can be observed that different algorithms have different abilities to maintain memory for different land cover types.For the ILT and LWF algorithms, the large black areas in the images indicate that they both classify most of the land cover features as background. In addition, it can be seen that the LWF algorithm performs better than the ILT algorithm in terms of road category prediction. The IoU for road recognition is 36.32% for LWF and 25.54% for ILT. From the images, it can also be observed that the predicted map from the LWF algorithm has better skeleton and connectivity for road regions compared to ILT. The LWF algorithm also shows better performance in recognizing lakes, achieving 73.76% IoU, while ILT only achieves 59.16%. In comparison to the previous two algorithms, the LWF-MC algorithm clearly has more green and purple areas. The green represents irrigated cropland, and the purple represents deciduous forest, both of which were learned in the first stage. This indicates that the LWF-MC algorithm has a good memory for these two classes, and it does not assign other new classes to these two land cover categories when compared to the ground truth. From the table, it can be seen that the LWF-MC algorithm achieves an IoU of 51.03% for irrigated cropland, while the ILT and LWF algorithms only have 2.13% and 2.01%, respectively. The same trend is observed for deciduous forest recognition, with the LWF-MC algorithm achieving 56.11% IoU, while ILT

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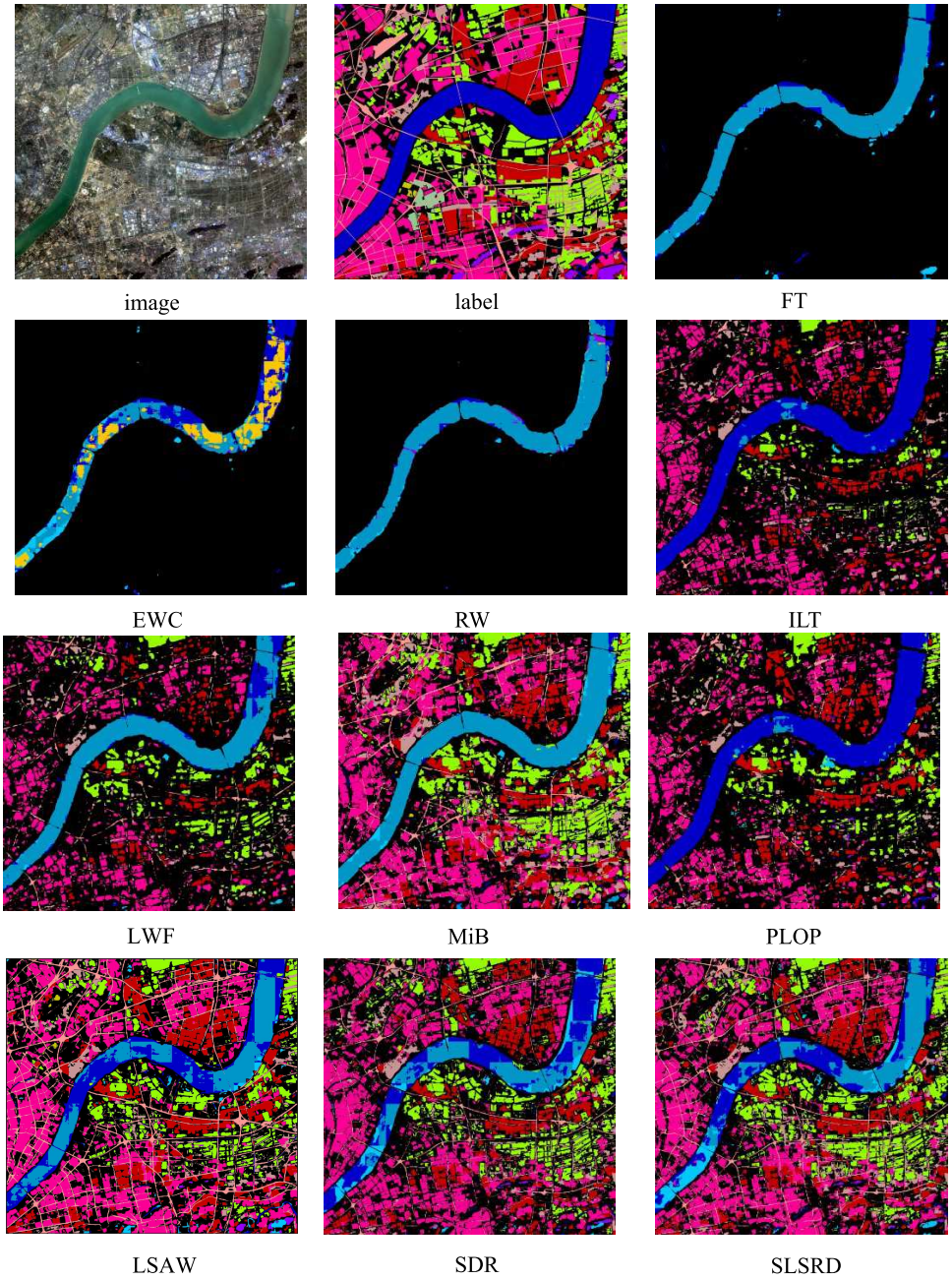


Fig. 9. Experimental results on the GID dataset (3-3-3-3-3).

and LWF algorithms only have 4.89% and 8.42%, respectively. Therefore, the predicted maps of the latter two algorithms are predominantly black, which is completely different from the results shown by the LWF-MC algorithm. As for the MiB and PLOP algorithms, it can be observed that the purple and green parts in the PLOP algorithm's predicted maps have spread outward, indicating that the PLOP algorithm incorrectly classified the new classes as irrigated cropland and deciduous forest classes learned in the first stage. On the other hand, the MiB algorithm does not exhibit this behavior, as the green and purple parts are roughly consistent with the ground truth, indicating that the MiB algorithm effectively learned the new classes while maintaining memory of the old classes. From the table, it can also be seen that the MiB algorithm achieves 71.61% and 69.48% IoU for irrigated cropland and deciduous forest, respectively, while PLOP achieves 70.92% and 66.42%. The MiB algorithm also achieves an overall mIoU of 33.25%, which is much better than the performance of the PLOP algorithm. For the LSAW algorithm, which is a recently proposed state-of-the-art method, it achieves a competitive overall mIoU of 32.85%. However, as shown in Table VI, it struggles with certain land cover types. For example, the LSAW algorithm achieves 42.43% IoU for industrial areas, which is lower than the 52.78% IoU achieved by the proposed SLSRD algorithm. Similarly, for roads, the LSAW algorithm achieves an IoU of 36.62%, while SLSRD achieves a significantly higher IoU of 52.16%. Although the LSAW algorithm shows notable performance in certain categories, such as arbor

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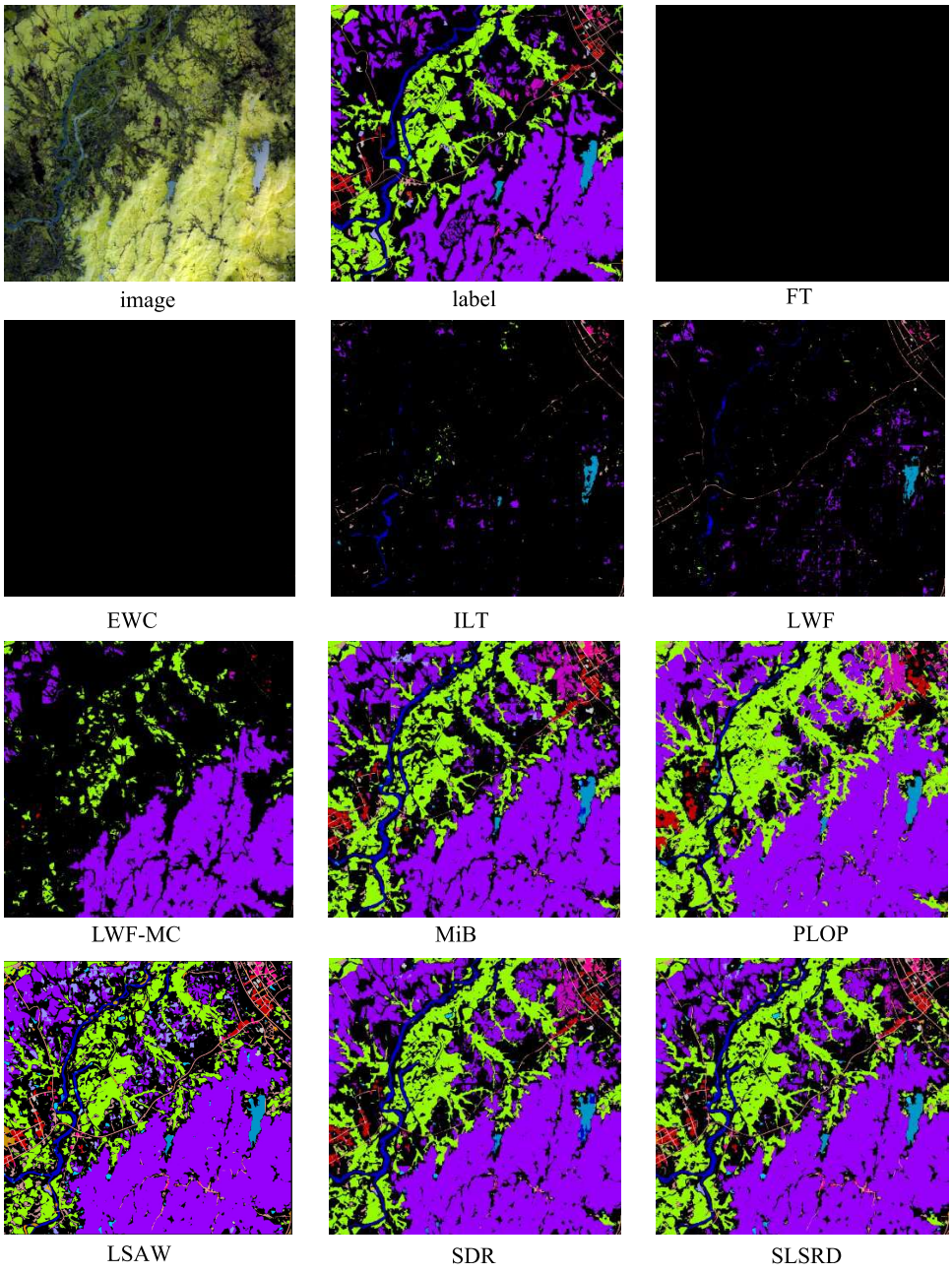


Fig. 10. Experimental results on the FBP dataset (6-3-3-3-3-3-3).

forest (73.49%), it still suffers from catastrophic forgetting in some other categories, which limits its overall performance. Compared to the aforementioned algorithms, the SLSRD algorithm proposed in this chapter achieves an mIoU of 42.57%. In addition, from the predicted maps, it can be observed that the algorithm maintains good road skeleton information, and the regions for irrigated cropland and deciduous forest are almost identical to those in the ground truth. Moreover, the algorithm is able to recognize the road passing through the deciduous forest in the lower right corner of the image, while other algorithms perform poorly in this area. As a result, the algorithm achieves an IoU of 52.16% for roads, which is significantly better than the IoU achieved by the LSAW algorithm and others.

C. Ablation Experiment

Using the MiB algorithm as the baseline for comparison, this section conducted ablation experiments on the Vaihingen, GID, and FBP datasets. Tables VII–IX represent the results of the ablation experiments on the Vaihingen, GID, and FBP datasets, respectively. Table X represents the results of the ablation experiment of the hyperparameters.

1) Analysis of the Ablation Experiment Results on the Vaihingen Dataset: First, the modified CE loss was added to the baseline. By incorporating pseudo-labels into the CE loss, the model's ability to distinguish between old and new classes was enhanced, making it more effective in learning new classes. Particularly, there was a significant improvement in the mIoU for the fifth class. Adding the proposed CE

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TABLE VII
RESULTS OF THE ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET

Algorithm	IoU						Acc						mIoU
	0	1	2	3	4	5	0	1	2	3	4	5	
	3-2												
MiB	81.1	84.6	55.4	65.9	18.8	62.8	91.0	92.6	63.6	73.0	22.2	80.6	61.4
MiB+CE	**84.4**	88.7	64.9	**69.9**	24.6	65.5	**93.6**	93.1	79.3	**78.3**	30.9	80.3	66.3
MiB+KD	81.2	86.6	59.9	63.7	24.9	64.3	93.1	91.6	69.9	68.2	30.5	83.0	63.4
MiB+CE+KD	84.3	**88.7**	66.7	67.0	**40.6**	**65.8**	93.0	**93.8**	**82.9**	72.5	**52.8**	**83.3**	**68.9**
	3-1-1												
MiB	74.0	84.9	56.0	60.2	25.6	27.3	93.2	**92.5**	67.2	**88.4**	31.4	27.9	54.7
MiB+CE	**82.0**	84.3	60.6	**67.2**	34.1	64.6	92.4	91.6	73.7	74.4	59.1	78.4	65.5
MiB+KD	80.5	**86.1**	60.4	65.0	26.3	62.7	92.3	91.8	72.0	69.9	45.4	79.4	63.5
MiB+CE+KD	80.9	85.5	**63.0**	64.9	**37.5**	**65.6**	**92.8**	91.4	**76.2**	69.0	**61.9**	**81.6**	**66.2**
	2-2-1												
MiB	74.0	84.9	56.0	60.2	25.6	27.3	93.2	92.5	67.2	**88.4**	31.4	27.9	54.7
MiB+CE	83.4	87.7	64.9	66.8	**37.8**	65.0	93.3	92.8	82.5	73.3	49.9	80.8	67.6
MiB+KD	78.6	86.7	62.1	62.8	27.5	61.1	92.2	91.5	73.9	67.6	54.4	76.8	63.1
MiB+CE+KD	**85.5**	**89.3**	**66.3**	**67.8**	34.1	**66.2**	**93.5**	**94.1**	**83.5**	74.7	**51.9**	**82.0**	**68.2**

TABLE VIII
RESULTS OF THE ABLATION EXPERIMENTS ON THE GID DATASET

	10-5			6-6-3				3-3-3-3-3		
Algorithm	0-10	11-15	mIoU	0-6	7-13	13-15	mIoU	0-9	10-15	mIoU
MiB	37.49	17.04	31.10	49.07	17.93	7.52	29.60	36.60	11.06	26.33
MiB+CE	39.31	29.96	36.79	52.12	15.25	11.95	30.76	40.56	12.09	29.88
MiB+KD	38.44	24.57	34.11	49.09	**19.70**	**18.66**	32.36	34.22	15.32	27.13
MiB+CE+KD	**40.42**	**30.53**	**37.33**	**53.87**	18.90	13.34	**33.16**	**37.5**	**19.51**	**30.75**

TABLE IX
RESULTS OF THE ABLATION EXPERIMENTS ON THE FBP DATASET

	19-5			16-8			8-8-8			
Algorithm	0-19	20-24	mIoU	0-15	16-24	mIoU	0-8	9-16	17-24	mIoU
MiB	50.13	15.02	37.49	47.43	16.34	37.48	48.18	44.47	16.2	36.76
MiB+CE	48.44	28.88	41.40	48.12	29.49	42.16	47.36	43.08	30.72	40.67
MiB+KD	**50.42**	14.22	37.39	46.71	19.73	38.08	48.72	43.26	20.83	38.05
MiB+CE+KD	48.46	**36.19**	**44.05**	**47.92**	**35.86**	**44.06**	**49.41**	**48.28**	**36.60**	**44.95**

to three different incremental learning tasks improved the performance compared to the previous baseline. The most notable improvement was observed in the 2-2-1 task, where the mIoU for the fifth class increased from 9.77% to 34.68%. In addition, the overall mIoU improved from 57.71% to 65.64%, showing a 13.7% enhancement.

Next, the KD loss was added to the baseline. Due to the KD loss, the model's ability to remember the old classes was enhanced, leading to an overall improvement in performance. After adding KD to the 3-1-1 task, the overall mIoU increased from 58.74% to 62.15%, a 5.8% improvement. The best results were obtained when both the proposed CE and KD losses were added, specifically in the 3-1-1 and 2-2-1 tasks. When only the CE loss function was added in the 3-2 task, the results were comparable. Therefore, these two losses are mutually beneficial, as they can improve each other's results and work together to accomplish the task. The effectiveness of the proposed method has been demonstrated through the ablation experiments.

TABLE X
RESULTS OF THE HYPERPARAMETERS ON THE VAIHINGEN DATASET

α	β	**mIoU-old**	**mIoU-new**	**mIoU**
0.5	30	68.6	57.3	64.6
1.0	10	66.4	58.7	64.3
1.0	**30**	**68.2**	**58.9**	**65.5**
1.0	50	68.6	58.0	65.1
2.0	30	66.7	58.8	64.8

2) Analysis of the Ablation Experiment Results on the GID Dataset: For the GID dataset, this chapter conducted ablation experiments with three incremental tasks. The task names have the same meaning as mentioned above. "10-5" represents the first stage training with the background class and ten new

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classes, followed by the second stage training with five new classes. The meanings of the remaining incremental tasks are not elaborated further.

Regarding the addition of the modified CE component, it enhanced the learning of new classes in all three incremental tasks, resulting in improved mIoU values for the new classes and an overall increase in mIoU. The most significant improvements were observed in the 10-5 incremental task, with approximately a 2% mIoU increase in the first stage and around a 13% mIoU increase in the second stage. The overall mIoU also improved by approximately 5.6%.

Regarding the addition of the modified KD component, the reinforcement of memory for old classes led to better learning of new classes. As a result, the overall mIoU for all three incremental tasks improved compared to using the MiB algorithm alone. In the 10-5 incremental task, there was an approximately 3% increase in mIoU. Combining both the CE and KD components is the SLSRD algorithm proposed in this chapter, which effectively learns new classes while maintaining memory of old classes. It achieved the best results across all tasks.

3) Analysis of the Ablation Experiment Results on the FBP Dataset: Regarding the addition of the modified CE component, it can be observed that there is an improvement compared to using the MiB algorithm alone, mainly reflected in the mIoU of the new classes. For the 15-9 task, after adding CE, the mIoU for the five new classes in the second stage increased from 15.02% to 28.88%. The old classes in the first stage did not experience a significant decrease, resulting in an overall mIoU of 41.40%, which is approximately a 4% improvement compared to using the MiB algorithm alone. Similar results were observed for the 16-8 and 8-8-8 tasks, where the addition of the CE module effectively improved the mIoU of the new classes.

Regarding the addition of the modified KD component, the distillation part helps the model better remember the old classes. This allows the model to distinguish between the features learned from the old classes and the features of the new classes in the current stage, leading to improved Acc. For the three incremental tasks in this section, adding the KD component improved the recognition Acc for the old classes in the 15-9 and 8-8-8 tasks. Although there was a slight decrease in Acc for the old classes in the 16-8 task, there was an improvement in the mIoU of the new classes, resulting in an overall improvement in mIoU. Compared to only adding one component, the SLSRD algorithm proposed in this chapter achieved the highest mIoU with the combined effects of both the CE and KD components.

4) Analysis of the Hyperparameters on Vaihingen Dataset: The hyperparameters α and β in 18 are primarily used to balance the ability to learn new knowledge (plasticity) and the ability to retain old knowledge (stability). In actual training, since SRD only operates on regions with fewer "uncertain pixels," the value of $\mathcal{L}_{\text{kd}}$ is typically much smaller than $\mathcal{L}_{\text{ce}}$. Without increasing $\mathcal{L}_{\text{kd}}$, the impact of distillation loss on gradient updates is significantly weakened. Therefore, we need to set β to a larger value to ensure that the two have comparable influence in optimization. When β is too small, the distillation constraint is weakened, the model's plasticity is enhanced, but forgetting is significantly exacerbated. For example, on the Vaihingen dataset (3-2), reducing β from 30 to 10 results in a decrease of approximately 1.8% percentage points in the mIoU for old classes and a decrease of approximately 1.2% in the overall mIoU. When β is too large, the model overemphasizes the retention of old knowledge, thereby inhibiting the learning of new classes. On the Vaihingen dataset (3-2), increasing β from 30 to 50 reduced the mIoU for new classes by approximately 0.9%, and the overall mIoU by approximately 0.4%. When $\beta = 30$, the influence of the SRD loss is effectively amplified, making it nearly as weighted as the CE loss in gradient updates, resulting in optimal overall performance. Regarding α, we found that a smaller α ($\alpha = 0.5$) weakened the model's supervisory signal for new classes, significantly reducing the mIoU for new classes. For example, on the Vaihingen dataset (3-2), when $\alpha = 0.5$ and $\beta = 30$, the mIoU for new classes decreased by approximately 1.6 % compared to $\alpha = 1$, and the overall mIoU decreased by approximately 0.9%. Larger α (such as $\alpha = 2$) allows the model to adapt more quickly to new classes, but this increases the likelihood of forgetting knowledge about old classes. On the Vaihingen dataset (3-2), the mIoU of the old class decreased by about 1.5%, and the overall mIoU decreased by about 0.7%. In summary, we choose the balance value $\alpha = 1$ and $\beta = 30$ to achieve the best balance between stability and plasticity, which can effectively learn new classes while well preserving knowledge of old classes.

V. Conclusion

We propose a novel method, SLSRD, to tackle incremental learning challenges in remote sensing image classification. The key innovation lies in using soft labels to replace traditional hard labels. During training, soft labels are used for uncertain pixels of old classes. These soft labels are derived from a combination of softmax and sigmoid functions. This approach helps mitigate catastrophic forgetting. It also retains essential information about old classes. In the KD phase, we introduce an efficient breadth-first search to extract subregions from the image, compute their weights, and calculate the similarity between the old and new model outputs. This process preserves recognition abilities for old classes while incorporating new knowledge. Extensive evaluations on benchmark datasets demonstrate significant improvements in both Acc and efficiency. In summary, the SLSRD method represents a substantial advancement in the field of incremental learning. It proficiently mitigates the issue of catastrophic forgetting while offering a computationally efficient approach for the incorporation of new classes. Furthermore, this method paves the way for future research avenues, particularly in the realms of multimodal data processing and the generalization of models across varied environments.

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REFERENCES

[1] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[2] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5168–5177.

[3] Z. Zhang, X. Zhang, C. Peng, X. Xue, and J. Sun, "ExFuse: Enhancing feature fusion for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 273–288.

[4] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[5] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 833–851.

[6] L.-C. Chen, Y. Yang, J. Wang, W. Xu, and A. L. Yuille, "Attention to scale: Scale-aware semantic image segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3640–3649.

[7] A. M. Kirillov et al., "Segment anything," in *Proc. IEEE Int. Conf. Comput. Vis.*, Jun. 2023, pp. 4015–4026.

[8] X. Lai et al., "LISA: Reasoning segmentation via large language model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 9579–9589.

[9] H. Zhang et al., "LLaVA-grounding: Grounded visual chat with large multimodal models," in *Proc. Eur. Conf. Comput. Vision.*, 2024, pp. 19–35.

[10] Z. Xia, D. Han, Y. Han, X. Pan, S. Song, and G. Huang, "GSVA: Generalized segmentation via multimodal large language models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3858–3869.

[11] Z. Ren et al., "PixelLM: Pixel reasoning with large multimodal model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26364–26373.

[12] J. He et al., "Multi-modal instruction tuned LLMs with fine-grained visual perception," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 13980–13990.

[13] M. McCloskey and N. J. Cohen, "Catastrophic interference in connectionist networks: The sequential learning problem," in *Psychology of Learning and Motivation*. Amsterdam, The Netherlands: Elsevier, 1989, pp. 109–165.

[14] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2018.

[15] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "ICaRL: Incremental classifier and representation learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5533–5542.

[16] S. Hou, X. Pan, C. C. Loy, Z. Wang, and D. Lin, "Learning a unified classifier incrementally via rebalancing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 831–839.

[17] K. Shmelkov, C. Schmid, and K. Alahari, "Incremental learning of object detectors without catastrophic forgetting," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3420–3429.

[18] J.-M. Perez-Rua, X. Zhu, T. M. Hospedales, and T. Xiang, "Incremental few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 13846–13855.

[19] K. J. Joseph, S. Khan, F. S. Khan, and V. N. Balasubramanian, "Towards open world object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5826–5836.

[20] V. Badrinarayanan, A. Handa, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for robust semantic pixel-wise labelling," 2015, *arXiv:1505.07293*.

[21] H. Noh, S. Hong, and B. Han, "Learning deconvolution network for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1520–1528.

[22] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.

[23] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Lect. Notes Comput. Sci.*, 2015, pp. 234–241.

[24] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking Atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[25] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[26] Y. Gao, M. Zhang, W. Li, X. Song, X. Jiang, and Y. Ma, "Adversarial complementary learning for multisource remote sensing classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505613.

[27] J. Wang, W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Hyperspectral and SAR image classification via multiscale interactive fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10823–10837, Dec. 2023.

[28] W. Li, J. Wang, Y. Gao, M. Zhang, R. Tao, and B. Zhang, "Graph-feature-enhanced selective assignment network for hyperspectral and multispectral data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526914.

[29] O. Ostapenko, M. Puscas, T. Klein, P. Jähnichen, and M. Nabi, "Learning to remember: A synaptic plasticity driven framework for continual learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11321–11329.

[30] R. Kemker and C. Kanan, "FearNet: Brain-inspired model for incremental learning," 2017, *arXiv:1711.10563*.

[31] Y. Xiang, Y. Fu, P. Ji, and H. Huang, "Incremental learning using conditional adversarial networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6618–6627.

[32] Y. Cong, M. Zhao, J. Li, S. Wang, and L. Carin, "GAN memory with no forgetting," in *Proc. Adv. neural inf. process. syst*, 2020, pp. 16481–16494.

[33] S. Cha, B. kim, Y. Yoo, and T. Moon, "SSUL: Semantic segmentation with unknown label for exemplar-based class-incremental learning," in *Proc. Adv. neural inf. process. syst*, 2021, pp. 10919–10930.

[34] Y. Oh, D. Baek, and B. Ham, "ALIFE: Adaptive logit regularizer and feature replay for incremental semantic segmentation," in *Proc. Adv. neural inf. process. syst*, 2022, pp. 14516–14528.

[35] L. Zhu, T. Chen, J. Yin, S. See, and J. Liu, "Continual semantic segmentation with automatic memory sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, Jun. 2023, pp. 3082–3092.

[36] A. Mallya and S. Lazebnik, "PackNet: Adding multiple tasks to a single network by iterative pruning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7765–7773.

[37] A. Mallya, D. Davis, and S. Lazebnik, "Piggyback: Adapting a single network to multiple tasks by learning to mask weights," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 67–82.

[38] A. A. Rusu et al., "Progressive neural networks," 2016, *arXiv:1606.04671*.

[39] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 3987–3995.

[40] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Natl. Acad. Sci.*, vol. 114, no. 13, pp. 3521–3526, 2017.

[41] A. Chaudhry, P. K. Dokania, T. Ajanthan, and P. H. S. Torr, "Riemannian walk for incremental learning: Understanding forgetting and intransigence," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 556–572.

[42] U. Michieli and P. Zanuttigh, "Incremental learning techniques for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 1–16.

[43] F. Cermelli, M. Mancini, S. R. Bulo, E. Ricci, and B. Caputo, "Modeling the background for incremental learning in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9230–9239.

[44] A. Douillard, Y. Chen, A. Dapogny, and M. Cord, "PLOP: Learning without forgetting for continual semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4040–4050.

[45] U. Michieli and P. Zanuttigh, "Continual semantic segmentation via repulsion-attraction of sparse and disentangled latent representations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1114–1124.

[46] L. Shan, W. Wang, K. Lv, and B. Luo, "Class-incremental learning for semantic segmentation in aerial imagery via distillation in all aspects," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615712.

[47] W. Huang, M. Ding, and F. Deng, "Domain-incremental learning for remote sensing semantic segmentation with multifeature constraints in graph space," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5645215.

[48] Y. Tai, Y. Tan, S. Xiong, and J. Tian, "Mine-distill-prototypes for complete few-shot class-incremental learning in image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5206013.

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[49] X. Lu, X. Sun, W. Diao, Y. Feng, P. Wang, and K. Fu, "LIL: Lightweight incremental learning approach through feature transfer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5611320.
[50] L. Bruzzone and D. F. Prieto, "An incremental-learning neural network for the classification of remote-sensing images," *Pattern Recognit. Lett.*, vol. 20, nos. 11–13, pp. 1241–1248, Nov. 1999.
[51] O. Tasar, Y. Tarabalka, and P. Alliez, "Incremental learning for semantic segmentation of large-scale remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3524–3537, Sep. 2019.
[52] X. Rong, X. Sun, W. Diao, P. Wang, Z. Yuan, and H. Wang, "Historical information-guided class-incremental semantic segmentation in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622618.
[53] *Potsdam and Vaihingen Dataset*. Accessed: Jan. 2024. [Online]. Available: https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/default.aspx
[54] S.-S. Kim and K.-C. Kwak, "Incremental modeling with rough and fine tuning method," *Appl. Soft Comput.*, vol. 11, no. 1, pp. 585–591, Jan. 2011.
[55] B. Ren et al., "Incremental land cover classification via label strategy and adaptive weights," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5624015.



Bo Ren (Member, IEEE) received the B.S. degree in telecommunications engineering from Northwest University, Xian, China, in 2011, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2017.

In 2019, he was a Visiting Scholar at Grenoble Images Parole Signal Automatique Laboratory (GIPSA-Lab), Grenoble, France. Since 2022, he has been an Associate Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. His research interests include data fusion, machine learning, and incremental learning in remote sensing images.



Zhao Wang received the B.S. degree from Xidian University, Xi'an, China, in 2021, where he is currently pursuing the M.S. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, School of Artificial Intelligence.

His research interests include deep learning, incremental learning, and land cover classification.



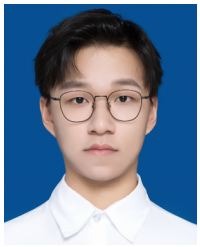
Hanyuan Ge received the B.S. degree from Xidian University, Xi'an, China, in 2024, where he is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, School of Artificial Intelligence.

His research interests include machine learning, incremental learning, and loss function design.



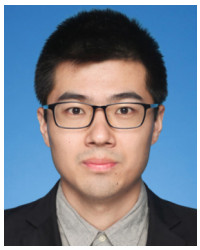
Biao Hou (Senior Member, IEEE) received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence, Xidian University, where he is currently a Professor. His research interests include deep learning and synthetic aperture radar image interpretation.



Bo Liu received the B.S. degree from Hefei University of Technology, Hefei, China, in 2022. He is currently pursuing the M.S. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include multimodal remote sensing image fusion, contrast learning, and land cover classification.



Chen Yang received the B.S. and Ph.D. degrees in circuits and systems from Xidian University, Xi'an, China, in 2016 and 2022, respectively.

Since 2022, he has been a Lecturer with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University. His research interests include intelligent interpretation of polarimetric SAR images, design of physical mechanism deep networks, and terrain element extraction of remote sensing images.



Licheng Jiao (Life Fellow, IEEE) was born in Shaanxi, China, in October 1959. He received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

From 1984 to 1986, he was an Assistant Professor with the Civil Aviation Institute of China, Tianjin, China. From 1990 to 1991, he was a Post-Doctoral Fellow with the Key Laboratory for Radar Signal Processing, Xidian University, Xi'an. Currently, he is the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. His research interests include signal and image processing, nonlinear circuits and systems theory, learning theory and algorithms, optimization problems, wavelet theory, and machine learning.

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